VECTOR BUNDLES WITH HOLOMORPHIC CONNECTIONS BUT NO FLAT HOLOMORPHIC CONNECTIONS

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Abstract. A holomorphic vector bundle on a compact Kähler manifold which admits a holomorphic connection has vanishing real Chern classes of positive degree by a theorem of Atiyah, and it is a long-standing question whether such a bundle admits a flat holomorphic connection. We construct a holomorphic vector bundle on a smooth projective surface which admits a holomorphic connection but no flat holomorphic connection, answering the question negatively. The associated projective bundle disproves a conjecture formulated by Höring on compact Kähler manifolds with split tangent bundle, in which the summands are not assumed to be integrable, and shows that the integrability assumption in Beauville's conjecture on such manifolds cannot be dropped even if one asks only for a product decomposition of the universal cover into factors whose dimensions are the ranks of the summands.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Holomorphic connections (see Definition 2.1) on holomorphic fiber bundles were introduced by Atiyah [Ati57], who applied them to a problem first studied by Weil [Wei38]: to characterize the holomorphic bundles which arise from representations of the fundamental group. A holomorphic bundle arises from a representation of the fundamental group if and only if it admits a flat holomorphic connection [Ati57, Proposition 14]. On a compact Riemann surface there are no non-zero holomorphic 2-forms, so any holomorphic connection is flat, and a holomorphic vector bundle admits a holomorphic connection if and only if it is a direct sum of indecomposable bundles of degree zero [Wei38], [Ati57, Theorems 2, 8, and 10]. See [HL19, Theorem 3.1] for the precise statement.

On a compact Kähler manifold of any dimension, a holomorphic vector bundle which admits a holomorphic connection has vanishing real Chern classes of positive degree [Ati57, Theorem 4, pp. 192–193], so the real Chern classes are no obstruction to a flat holomorphic connection on such a bundle. On the other hand, the criterion of Weil does not extend to higher dimensions: on the product of $\mathbb{P}^1$ and an elliptic curve there exists an indecomposable vector bundle of rank two

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with vanishing Chern classes which admits no holomorphic connection [Ati57, Proposition 22]. In dimension at least two a holomorphic connection need not be flat, and the following question has remained open.

Question 1.1. Let X be a compact Kähler manifold and E a holomorphic vector bundle on X which admits a holomorphic connection. Does E admit a flat holomorphic connection?

In [Bis95, §1], Question 1.1 is described as well known, and in [BI07, §1] it is posed for compact complex manifolds and described as long-standing. In [BD20, §1] and [BD24, §1], Question 1.1 is attributed to Atiyah. Question 1.1 is not posed explicitly in [Ati57], whose last section discusses problems on connections over bases of dimension greater than one [Ati57, §8].

For principal bundles whose structure group is a compact complex torus, which is not a linear group, the analogue of Question 1.1 has a negative answer, so such counterexamples do not concern vector bundles. Such principal bundles over complex tori that admit a holomorphic connection but no flat holomorphic connection were constructed in [Mur59, Mur61], and [BD20, Proposition 4.3] constructs a principal bundle over a projective K3 surface, with an elliptic curve as structure group, which admits a holomorphic connection but no flat holomorphic connection.

We now list known cases in which Question 1.1 has a positive answer. Let X and E be as in Question 1.1. Since Question 1.1 for X reduces to Question 1.1 for each connected component of X, we assume that X is connected.

(1) Fix a Kähler form on X, and let $\mu_{\min}(T_X)$ be the minimal slope of the graded pieces of the Harder–Narasimhan filtration of T_X with respect to the Kähler form. If $\mu_{\min}(T_X) > 0$, then E admits a flat holomorphic connection and is a direct sum of stable bundles of slope zero [Bis95, Theorem 3.1]. If X is projective and $\deg T_X \geq 0$ with respect to an ample line bundle L on X, then E admits a flat holomorphic connection provided that either the last graded piece of the Harder–Narasimhan filtration of T_X with respect to L has non-negative degree, or T_X is not semistable, the maximal semistable subsheaf of T_X is locally free, and the Néron–Severi group of X has rank one [Bis98, Theorem 2.4].
(2) If a finite étale cover of X admits a holomorphic submersion onto a complex torus whose fibers are rationally connected, then E admits a flat holomorphic connection [BI07, Theorem 1.1].
(3) If X is simply connected and $c_1(T_X) = 0$, then E is trivial [BD19, Theorem 6.2]. More generally, if $c_1(T_X) = 0$ in $H^2(X, \mathbb{R})$, then X carries a Ricci-flat Kähler form by Yau's theorem (cf. [BD24, §2]), so E admits a flat holomorphic connection [BD24, Theorem 2.3].
(4) On a projective manifold, any vector bundle which is semistable with respect to an ample line bundle and has vanishing Chern classes admits a flat holomorphic connection, and it is an extension of unitary flat bundles [Sim92, Corollary 3.10 and the paragraph after it]. Since the real Chern classes of E of positive degree vanish by [Ati57, Theorem 4, pp. 192–193], it follows that, when X is projective, the bundle E in a counterexample to Question 1.1 is not semistable with respect to any ample line bundle. On the other hand, a vector bundle which is not semistable may still admit a flat holomorphic connection [Bis95, Remark 3.7(iii)].
(5) If E has a filtration by holomorphic subbundles whose graded pieces are isomorphic to $\mathcal{O}_X$, then, fixing isomorphisms between the graded pieces and $\mathcal{O}_X$, E is associated with a principal bundle whose structure group is the group of upper triangular unipotent matrices, and E admits a flat holomorphic connection [BG08, Corollary 3.8 and the paragraph before Corollary 3.9].

An extension of two holomorphic vector bundles which admit holomorphic connections need not admit a holomorphic connection: on any compact Riemann surface of genus at least two, [BH15, §§1–2] constructs such an extension which admits no holomorphic connection.

We now turn to the second problem of this paper. Let X be a compact Kähler manifold and $\pi : \widetilde{X} \to X$ the universal cover. If $\widetilde{X}$ is a product $Y_1 \times \cdots \times Y_k$ of complex manifolds on which

$\pi_1(X)$ acts diagonally, then the splitting

$$T_{\widetilde{X}} = \bigoplus_{i=1}^{k} \mathrm{pr}_i^* \, T_{Y_i}$$

descends to a splitting of T_X into holomorphic subbundles, where pr_i is the projection onto Y_i. Beauville asked whether, conversely, any splitting of T_X into holomorphic subbundles arises in this way [Bea00, Introduction]. He gave a positive answer when X admits a Kähler–Einstein metric, when T_X is a direct sum of line bundles of negative degree with respect to a Kähler class, and when $\dim X = 2$ [Bea00, Theorems A, B, and C], and showed by Hopf manifolds that the Kähler assumption cannot be dropped [Bea00, (2.1)].

The subbundles of T_X which arise from a product decomposition of $\widetilde{X}$, and their direct sums, are integrable. On the other hand, if A is an abelian surface, then $T_{A\times\mathbb{P}^1}$ is the direct sum of the pullback of $T_{\mathbb{P}^1}$ and a subbundle of rank two which is not integrable [Bea00, (2.2)]. Beauville therefore conjectured the following [Bea00, (2.3)] (see also [Hör26, Conjecture 1.4]).

Conjecture 1.2 (Beauville)**.** *Let X be a compact Kähler manifold such that $T_X = V_1 \oplus \cdots \oplus V_k$ for holomorphic subbundles $V_1, \ldots, V_k \subset T_X$, and let $\pi : \widetilde{X} \to X$ be the universal cover. Assume that $\bigoplus_{i\in J} V_i$ is integrable for any subset $J \subset \{1, \ldots, k\}$. Then there exist complex manifolds $Y_1, \ldots, Y_k$ and a biholomorphism $\Phi : \widetilde{X} \to Y_1 \times \cdots \times Y_k$ such that $d\Phi(\pi^* V_i) = \Phi^* \, \mathrm{pr}_i^* \, T_{Y_i}$ as subbundles of $\Phi^* T_{Y_1\times\cdots\times Y_k}$ for $i = 1, \ldots, k$.*

For $k = 2$, Conjecture 1.2 is proved in [OAI26b, Theorem 1.1]. Earlier partial results include [Dru00, Theorems 1 and 2], [CP02, Corollary 3.7 and Theorem 4.1], [BPT06, Theorem 1.1(i)], [PT13, Theorem B], [DPPT22, Theorem 1.4], and [DPPT24, Theorem 1.1].

Without any assumption on integrability, Campana and Peternell asked whether the universal cover of a projective manifold X with $T_X = \bigoplus_{i=1}^k E_i$ is a product $\prod_{i=1}^k A_i$ of complex manifolds, and, when the E_i are integrable, whether the splitting of T_X lifts to such a product [CP02, Introduction]. In the example of [Bea00, (2.2)] (see also [Hör07, Example 2.14]), whose splitting has a non-integrable summand, the universal cover is still a product of complex manifolds whose dimensions are the ranks of the summands [Hör07, §1]. Höring proposed the following conjecture, in which no integrability is assumed and the dimensions of the factors are the ranks of the summands [Hör07, Conjecture 1.1].

Conjecture 1.3 ([Hör07, Conjecture 1.1])**.** *Let X be a connected compact Kähler manifold such that $T_X = V_1 \oplus V_2$ for holomorphic subbundles $V_1, V_2 \subset T_X$ of positive rank, and let $\pi : \widetilde{X} \to X$ be the universal cover. Then there exist complex manifolds Y_1, Y_2 with $\dim Y_j = \operatorname{rank} V_j$ for $j = 1, 2$ and a biholomorphism $\Phi : \widetilde{X} \to Y_1 \times Y_2$. If moreover V_j is integrable for some j, then there exists such a biholomorphism Φ with $d\Phi(\pi^* V_j) = \Phi^* \, \mathrm{pr}_j^* \, T_{Y_j}$.*

In [Hör08, Conjecture 1.6], the condition $\dim Y_j = \operatorname{rank} V_j$ is replaced by isomorphisms of vector bundles $\pi^* V_j \cong \Phi^* \, \mathrm{pr}_j^* \, T_{Y_j}$ for $j = 1, 2$. Höring describes Conjecture 1.3 as a conjecture initially asked by Beauville [Hör07, §1], and labels [Hör08, Conjecture 1.6] as a conjecture of Beauville [Hör08, §1]. Unlike Conjecture 1.2, they assume no integrability, and they ask for compatibility with the splitting, as subbundles, only for integrable summands.

Conjecture 1.3 holds when V_1 and V_2 are both integrable [OAI26b, Theorem 1.1]. The summands V_1 and V_2 are both integrable when X is projective and not uniruled [Hör07, Theorem 1.3], and when X is a rationally connected projective manifold [OAI26a, Theorem 1.1], as conjectured by Höring [Hör08, Conjecture 1.2], [Hör26, Conjecture 1.5]. Thus Conjecture 1.3 holds in both cases [OAI26b, Corollary 1.2], [OAI26a, Corollary 1.2]. In the rationally connected case, X is a product compatible with the splitting as soon as one of the summands is integrable [Hör07, Theorem 1.4]. Thus one of the summands in a counterexample to Conjecture 1.3 is not integrable, and a projective counterexample is uniruled and not rationally connected. There are also partial results in the uniruled case [Hör08, Proposition 3.16, Theorems 1.3 and 1.7]. For a

$\mathbb{Q}$-factorial klt projective variety X of Fano type with $T_X = V_1 \oplus V_2$, the summands V_1 and V_2 are integrable with algebraic leaves [Hör26, Theorem 1.1], and for any normal projective variety X of Fano type with $T_X = V_1 \oplus V_2$, there exists a product decomposition compatible with the splitting after a finite cover which is étale in codimension one [Hör26, Corollary 1.2].

Splittings of the tangent bundle also arise from fiber bundles. Let $p : X \to Y$ be a holomorphic submersion of compact complex manifolds with a holomorphic subbundle $H \subset T_X$ such that $T_X = T_{X/Y} \oplus H$, where $T_{X/Y} := \ker(dp)$ is the relative tangent bundle. If H is integrable, then the universal cover of X is a product by the Ehresmann theorem [CL85, Chapter V, §2, Proposition 1 and §4, Theorem 3], see also [Hör07, Theorem 3.17]. The integrability of H is proved in [Hör07, Lemma 3.18] when X is projective and Y is a rationally connected projective manifold, and then $X \cong Y \times F$ for a fiber F of p.

Let $p : X \to Y$ be a projective holomorphic fiber bundle whose fiber F is a normal projective variety with $\mathrm{Pic}^0(F) = 0$, over a projective variety Y with log terminal singularities and numerically trivial canonical class. If the relative tangent sequence of p splits holomorphically, then p admits locally constant transition functions [Mül25, Lemma 4.1]. For projective bundles over a projective manifold of any dimension, [CP02, Theorem 2.7(1)] states that if a projective manifold X admits a $\mathbb{P}^r$-bundle structure $X \to Y$ whose relative tangent sequence splits, then the universal cover of X is isomorphic to $\widetilde{Y} \times \mathbb{P}^r$, where $\widetilde{Y}$ is the universal cover of Y. The universal cover of the $\mathbb{P}^{261}$-bundle of Corollary 1.5 is not biholomorphic to the product of $\mathbb{P}^{261}$ and the universal cover of the base, and we compare Corollary 1.5 with [CP02, Theorem 2.7(1)] and the proof given there in Remark 7.1.

We now explain how Question 1.1 and Conjecture 1.3 are related. Let Z be a connected compact Kähler manifold with $\dim Z > 0$, let E be a holomorphic vector bundle on Z of rank $n+1 \geq 2$ with a holomorphic connection ∇, and let $\mathbb{P}(E) \to Z$ be the projective bundle of lines in E. Then ∇ defines a holomorphic subbundle $H_\nabla \subset T_{\mathbb{P}(E)}$ with

$$T_{\mathbb{P}(E)} = T_{\mathbb{P}(E)/Z} \oplus H_\nabla$$

(see Definition-Lemma 6.1). Since $\mathbb{P}(E)$ is a compact Kähler manifold [Voi02, Proposition 3.18], which is connected because Z and $\mathbb{P}^n$ are connected, Conjecture 1.3 predicts a product decomposition of the universal cover of $\mathbb{P}(E)$ with $V_1 = T_{\mathbb{P}(E)/Z}$ and $V_2 = H_\nabla$. If H_∇ is integrable, then the universal cover of $\mathbb{P}(E)$ is a product by the Ehresmann theorem (see [Hör07, Theorem 3.17]). In general, by Proposition 6.3, whose condition (1) requires Y_1 and Y_2 to be connected and simply connected, which holds because complex manifolds Y_1, Y_2 with $Y_1 \times Y_2$ biholomorphic to a connected simply connected manifold are connected, as images of the projections, hence path-connected, and simply connected [Hat02, Proposition 1.12], the first part of [Hör08, Conjecture 1.6] for $V_1 = T_{\mathbb{P}(E)/Z}$ and $V_2 = H_\nabla$ holds if and only if the pullback of $\mathbb{P}(E)$ to the universal cover $\widetilde{Z}$ of Z is isomorphic to $\widetilde{Z} \times \mathbb{P}^n$ over $\widetilde{Z}$. Here the hypotheses of Proposition 6.3 on the fiber hold, since $\mathbb{P}^n$ is a compact, connected, and simply connected complex manifold of dimension $n \geq 1$ [Hat02, Example 0.6 and Proposition 1.26] with $H^0(\mathbb{P}^n, \Omega^1_{\mathbb{P}^n}) = 0$ [GH78, Chapter 0, §7, p. 118].

Now apply Question 1.1 to E on Z. The projective form of Question 1.1 for E asks whether $\mathbb{P}(E)$ admits a flat holomorphic connection (see Definition 2.2). If $\mathbb{P}(E)$ admits such a connection, then so does the pullback of $\mathbb{P}(E)$ to $\widetilde{Z}$, since the pulled-back transition maps are still locally constant, and hence the pullback is isomorphic to $\widetilde{Z} \times \mathbb{P}^n$ over $\widetilde{Z}$ by Lemma 2.3(2). Thus a positive answer to the projective form of Question 1.1 for E implies the first parts of [Hör08, Conjecture 1.6] and of Conjecture 1.3 for $\mathbb{P}(E)$, and a counterexample to Conjecture 1.3 may be sought among projective bundles with holomorphic connections.

Since a projective counterexample to Conjecture 1.3 is uniruled and not rationally connected, as noted above, a natural choice is a projective bundle $\mathbb{P}(E)$ over a base Z which is not uniruled: then $\mathbb{P}(E)$ is uniruled, it is not rationally connected since Z is not uniruled, and H_∇ may be non-integrable. For such $\mathbb{P}(E)$, the splitting $T_{\mathbb{P}(E)} = T_{\mathbb{P}(E)/Z} \oplus H_\nabla$ is a counterexample to the

first part of [Hör08, Conjecture 1.6] if and only if the pullback of $\mathbb{P}(E)$ to $\widetilde{Z}$ is not isomorphic to $\widetilde{Z} \times \mathbb{P}^n$ over $\widetilde{Z}$, and then, by Lemma 2.3(2), the projective form of Question 1.1 has a negative answer for E.

1.1. **Main results.** In this paper, we answer Question 1.1 negatively. For a holomorphic vector bundle E, we denote by $\mathbb{P}(E)$ the holomorphic projective bundle of lines in E (see Definition 2.2). For holomorphic vector bundles N and Q, an extension of Q by N is a holomorphic vector bundle G with an exact sequence $0 \to N \to G \to Q \to 0$ (see Definition 2.4). Below we use the Hesse pencil of plane cubics [AD09], whose equation is given in Construction 3.2.

Theorem 1.4. *There exist a smooth projective curve C of genus two, a smooth projective surface S, a surjective morphism $f : S \to C$ with connected fibers such that $f_* : \pi_1(S) \to \pi_1(C)$ is an isomorphism, a holomorphic vector bundle M of rank 261 on C with a flat holomorphic connection ∇_M which induces the trivial connection on $\det M \cong \mathcal{O}_C$, and a holomorphic vector bundle E of rank 262 on S which fits into an exact sequence of holomorphic vector bundles*

$$0 \to f^*M \to E \to \mathcal{O}_S \to 0,$$

such that the following hold, where $q : \widetilde{S} \to S$ is the universal cover of S.

1. *E admits a holomorphic connection ∇ which preserves f^*M, induces $f^*\nabla_M$ on f^*M and the trivial connection d on $\mathcal{O}_S$, and has non-zero curvature.*
2. *E admits no flat holomorphic connection, and $\mathbb{P}(E)$ admits no flat holomorphic connection (see Definition 2.2).*
3. *The smooth complex vector bundle underlying q^*E is trivial, and $c_i(E) = 0$ in $H^{2i}(S, \mathbb{Z})$ for any $i \geq 1$.*
4. *There exists a smooth fiber F of f such that F is an elliptic curve, $E|_F$ is a non-split extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus 261}$, and $\mathbb{P}(E|_F)$ is not isomorphic to $F \times \mathbb{P}^{261}$ over F.*
5. *$\mathbb{P}(q^*E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$.*

Moreover, one can take S to be the base change of the Hesse pencil (see Construction 3.2) along a morphism $h : C \to \mathbb{P}^1$ of degree five none of whose branch values is one of the four points $\infty, -3, -3\epsilon, -3\epsilon^2$ of $\mathbb{P}^1$ in the coordinate of Construction 3.2, where ϵ is a primitive cube root of unity, and $M = \operatorname{Sym}^{260} E_0$ for a non-split extension E_0 of $\mathcal{L}^{-1}$ by $\mathcal{L}$, where $\mathcal{L}$ is a line bundle on C with $\mathcal{L}^{\otimes 2} \cong \omega_C$. In this case S is a smooth member of the linear system $|\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes h^*\mathcal{O}_{\mathbb{P}^1}(1)|$ on $\mathbb{P}^2 \times C$.

Parts (1) and (2) of Theorem 1.4 answer Question 1.1 negatively, since S is projective and thus a compact Kähler manifold [GH78, Chapter 0, §7]. Part (3) shows that E has no topological obstruction to a flat holomorphic connection (as smooth complex vector bundles, $E \cong f^*M \oplus \mathcal{O}_S$, and $f^*\nabla_M \oplus d$ is a flat connection on $f^*M \oplus \mathcal{O}_S$, see Section 7): the smooth complex vector bundle underlying q^*E is trivial, whereas by (5) the holomorphic $\mathbb{P}^{261}$-bundle $\mathbb{P}(q^*E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$.

Let S be the surface of Construction 3.2. Then S and $M = \operatorname{Sym}^{260} E_0$ are the explicit choices in the last part of Theorem 1.4, and for any φ and η as in Construction-Proposition 4.5, the holomorphic vector bundle E with the holomorphic connection ∇ constructed there satisfies parts (1)–(5) of Theorem 1.4. From Theorem 1.4 and the results of Section 6, we obtain the following for the projective bundle $\mathbb{P}(E)$ over S.

Corollary 1.5. *There exist a smooth projective variety X of dimension 263 and a decomposition $T_X = V_1 \oplus V_2$ into holomorphic subbundles with $\operatorname{rank} V_1 = 261$ and $\operatorname{rank} V_2 = 2$ such that V_1 is integrable, V_2 is not integrable, and the universal cover of X is not biholomorphic to $Y_1 \times Y_2$ for any complex manifolds Y_1, Y_2 with $\dim Y_1 = 261$ and $\dim Y_2 = 2$. More precisely, for any holomorphic vector bundle E on S with the holomorphic connection ∇ given by Construction-Proposition 4.5, one can take $X = \mathbb{P}(E)$, $V_1 = T_{X/S}$, and $V_2 = H_\nabla$ (see Definition-Lemma 6.1). In particular, Conjecture 1.3 and* [Hör08, Conjecture 1.6] *are false.*

Since V_2 is not integrable, the variety X of Corollary 1.5 does not satisfy the hypothesis of Conjecture 1.2. Thus Corollary 1.5 shows that the integrability assumption in Conjecture 1.2 cannot be dropped even if one asks only for a product decomposition of the universal cover into factors whose dimensions are the ranks of the summands, and that it cannot be weakened to the integrability of one of the summands.

Sketch of the proof. The proofs of Theorem 1.4 and Corollary 1.5 have three steps.

(1) Proposition 4.4 yields a non-zero $\varphi \in H^0(S, \Omega_S^2 \otimes f^*M)$ which is exact for $f^*\nabla_M$, and Construction-Proposition 4.5 turns φ and a primitive into E and ∇ with curvature $\left(\begin{smallmatrix} 0 & \varphi \\ 0 & 0 \end{smallmatrix}\right)$.
(2) By Corollary 3.7, the non-zero curvature makes $E|_F$ a non-split extension for some smooth fiber F of f, which lifts isomorphically to $\widetilde{S}$. Thus $\mathbb{P}(q^*E)$ is non-trivial over $\widetilde{S}$ (Proposition 5.2).
(3) Pulled back to the simply connected $\widetilde{S}$, a flat holomorphic connection on E or $\mathbb{P}(E)$ would trivialize $\mathbb{P}(q^*E)$ (Lemma 2.3). For Corollary 1.5, the second projection of any biholomorphism of the universal cover of $\mathbb{P}(E)$ onto a product of complex manifolds of dimensions 261 and 2 is constant on the fibers $\mathbb{P}^{261}$ by a dimension count, and Proposition 6.3 turns such a biholomorphism into such a trivialization.

The key new step is the existence of φ: no such form exists for unitary flat coefficients. Since E_0 is not semistable, the proof of Lemma 4.2 bounds $h^0(S, \Omega_S^2 \otimes f^*M)$ below by the quadratic function $(m+1)(m+6)$ of m for $M = \operatorname{Sym}^{2m} E_0$, while Lemma 4.3 bounds $\dim H^2$ of the local system of flat sections of f^*M by a linear one:

$$h^0(S, \Omega_S^2 \otimes f^*M) \geq (m+1)(m+6) = 17816 > 17748 = 68(2m+1) \quad \text{for } m = 130.$$

Thus φ is any non-zero element of the kernel of the map $H^0(S, \Omega_S^2 \otimes f^*M) \to H^2$ of that local system, $\psi \to [\psi]$, a kernel of dimension at least $17816 - 17748 = 68$, and φ is not computed. **Step 1** of the proof of Proposition 5.2, **Steps 2** and **3** of the proof of Proposition 6.3, and the proof of Corollary 1.5 in Section 7 are also new. The other links use the Lefschetz hyperplane theorem [Laz04, Theorem 3.1.21], the Atiyah–Weil criterion [Wei38], [Ati57, Theorems 2, 8, and 10], the Leray spectral sequence, and GAGA [Ser56] directly.

(a) Unstable flat coefficients. By item (4) of the list of known cases after Question 1.1, a counterexample to Question 1.1 on a projective manifold is not semistable with respect to any ample line bundle. The flat coefficients cannot be unitary: if $f^*\nabla_M$ preserved a Hermitian metric $\langle , \rangle$ and $\varphi = d_{f^*\nabla_M}\eta$, then the 4-form $\langle \varphi \wedge \bar{\varphi} \rangle$, positive where $\varphi \neq 0$, would equal $d\langle \eta \wedge \bar{\varphi} \rangle$, whose integral over the compact surface S is 0 by Stokes' theorem, so $\varphi = 0$.

We therefore build M from the bundle E_0 of Lemma 2.5, which has a flat holomorphic connection but contains $\mathcal{L}$ with $\deg \mathcal{L} = 1$. The graded pieces $\mathcal{L}^{\otimes(2m-2j)}$ of $\operatorname{Sym}^{2m} E_0$ with $0 \leq j < m$ have degree $2m-2j > 0$ and account for the quadratic part $m(m+1)$ of $(m+1)(m+6) = m(m+1) + 6(m+1)$, while the upper bound $68(2m+1)$ depends only on $e(S) = 60$ and on the four generators of $\pi_1(S) \cong \pi_1(C)$. The least m with $(m+1)(m+6) > 68(2m+1)$ is 130, so $\operatorname{rank} M = 261$. Any non-zero exact form can be used, since an upper triangular connection with flat diagonal blocks has curvature linear in the off-diagonal entry (Construction-Proposition 4.5).

(b) An obstruction on a compact curve. The holomorphic connections on E are $\nabla + \theta$ with $\theta \in H^0(S, \Omega_S^1 \otimes \operatorname{End} E)$, and flatness is a quadratic equation in θ. Instead of solving it, we use one smooth fiber F of f to exclude all flat holomorphic connections on E and on $\mathbb{P}(E)$ at once. Since S is an ample divisor in $\mathbb{P}^2 \times C$, the Lefschetz hyperplane theorem implies that $f_* : \pi_1(S) \to \pi_1(C)$ is an isomorphism, so loops in F are null-homotopic in S and F lifts isomorphically to $\widetilde{S}$ (Lemma 3.4). A flat connection would thus trivialize $\mathbb{P}(E|_F)$, while $\mathbb{P}(E|_F)$ is non-trivial when $E|_F$ is non-split, since F is compact (Lemma 5.1). In that case the flat connection $\nabla|_F$ has non-trivial monodromy along some loop in F which is null-homotopic in S, and this is possible only because ∇ has non-zero curvature.

(c) Descent of connections. Non-zero curvature alone does not prevent an extension from being pulled back from C: on the product of C and an elliptic curve $\mathbb{C}/\Lambda$, the split extension $\mathrm{pr}_1^*(M \oplus \mathcal{O}_C)$ carries holomorphic connections with non-zero curvature built from the 1-form $\mathrm{pr}_2^* dz$, which is not pulled back from C. **Step 1** of the proof of Proposition 5.2 argues by contradiction: if the image of the extension class in $H^0(C, R^1 f_* f^* M)$ vanished, then E would be pulled back from C as an extension, and ∇ would be flat by Corollary 3.7. Corollary 3.7 depends on Proposition 3.6, which uses that the j-invariant of the fibers of f is non-constant. By base change, the restriction of E to the fiber over c is non-split whenever the image does not vanish at c, which holds for all but finitely many $c \in C$.

(d) Rigidity of products. The product decompositions of [Hör07, Theorem 3.17] and [OAI26b, Theorem 1.1] depend on the integrability of the summands, while the horizontal subbundle H_∇ of Definition-Lemma 6.1 is not integrable (Lemma 6.2). Instead, we use that the fibers of $\mathbb{P}(q^*E) \to \widetilde{S}$ are copies of $\mathbb{P}^{261}$. Let Φ be a biholomorphism of $\mathbb{P}(q^*E)$ onto $Y_1 \times Y_2$ with $\dim Y_2 = 2$. On a fiber $\mathbb{P}^{261}$, the dimension of any non-empty fiber of $\mathrm{pr}_2 \circ \Phi$ is at least 259, and any two closed subvarieties of $\mathbb{P}^{261}$ of dimension at least 259 intersect, so $\mathrm{pr}_2 \circ \Phi$ is constant on $\mathbb{P}^{261}$. Thus Φ maps any fiber of $\mathbb{P}(q^*E) \to \widetilde{S}$ onto a slice $Y_1 \times \{y\}$, and Proposition 6.3 turns Φ into a trivialization of $\mathbb{P}(q^*E)$ over $\widetilde{S}$. No integrability of H_∇ or of the summands enters.

For the isomorphisms of vector bundles in [Hör08, Conjecture 1.6], Proposition 6.3 covers any compact, connected, and simply connected fiber F of positive dimension with $H^0(F, \Omega^1_F) = 0$: the isomorphisms $\pi^* H_\nabla \cong \Phi^* \mathrm{pr}_2^* T_{Y_2}$ and $H_\nabla \cong p^* T_S$ make the pullback of T_{Y_2} to F trivial, so the differential of $\mathrm{pr}_2 \circ \Phi$ on F consists of two holomorphic 1-forms and vanishes. This step uses that the fibers carry no non-zero holomorphic 1-form: for a product of two elliptic curves with the product splitting, the fibers $\{z\} \times \mathbb{C}$ of the universal cover $\mathbb{C}^2$ over the first factor carry the 1-form dw, and the shear $(z, w) \to (w, z + w)$ satisfies both isomorphisms but does not map the lines $\{z\} \times \mathbb{C}$ into slices.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce flat holomorphic connections on projective bundles, and recall that a non-split extension of $\mathcal{L}^{-1}$ by $\mathcal{L}$, where $\mathcal{L}$ is a line bundle with $\mathcal{L}^{\otimes 2}$ isomorphic to the canonical bundle of a smooth projective curve of genus at least two, admits a flat holomorphic connection.

In Section 3 we construct the elliptic surface S as the base change of the Hesse pencil along a morphism $h : C \to \mathbb{P}^1$ of degree five none of whose branch values is the parameter of a singular member of the Hesse pencil, prove that $f_* : \pi_1(S) \to \pi_1(C)$ is an isomorphism, and prove that, for any holomorphic vector bundle N on C, any holomorphic connection on f^*N is the pullback of a holomorphic connection on N.

In Section 4 we show, by a comparison of dimensions, that some non-zero holomorphic 2-form on S with values in f^*M is exact with respect to $f^*\nabla_M$, and we construct from such a form the extension E and the holomorphic connection ∇ with non-zero curvature.

In Section 5 we prove that the restriction of E to some smooth fiber of f is a non-split extension, and deduce that $\mathbb{P}(q^*E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$.

In Section 6 we prove that the horizontal subbundle of a holomorphic connection is not integrable when some value of the curvature is not a scalar multiple of the identity (Lemma 6.2), and we prove a criterion for a product decomposition of the universal cover of a fiber bundle in terms of the pullback of the fiber bundle to the universal cover of the base (Proposition 6.3).

In Section 7 we prove Theorem 1.4 and Corollary 1.5, and in Remark 7.1 we compare Corollary 1.5 with [CP02, Theorem 2.7] and the proof given there.

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2. Preliminaries

2.1. **Holomorphic connections.** Holomorphic connections were introduced in [Ati57, p. 188] as splittings of the Atiyah sequence. We use the following equivalent definition (cf. [Ati57, §4]).

Definition 2.1. Let X be a complex manifold and E a holomorphic vector bundle on X. A *holomorphic connection* on E is a $\mathbb{C}$-linear morphism of sheaves $\nabla : E \to \Omega^1_X \otimes E$ such that

$$\nabla(us) = du \otimes s + u\nabla s$$

for any local holomorphic function u and any local holomorphic section s of E. Extending ∇ to E-valued holomorphic forms by $\nabla(\omega \otimes s) = d\omega \otimes s + (-1)^k \omega \wedge \nabla s$ for a k-form ω, the *curvature* of ∇ is the section $R_\nabla \in H^0(X, \Omega^2_X \otimes \operatorname{End} E)$ ($\nabla \circ \nabla$ is $\mathcal{O}_X$-linear by the Leibniz rule) with $R_\nabla(s) = \nabla(\nabla s)$, and ∇ is *flat* if $R_\nabla = 0$. We write $d_\nabla := \nabla + \bar\partial_E$ for the smooth connection on the smooth complex vector bundle underlying E defined by ∇, extended to smooth E-valued forms.

If $E' \subset E$ is a holomorphic subbundle with $\nabla(E') \subset \Omega^1_X \otimes E'$, then we say that ∇ *preserves* E'. In this case, the holomorphic connections on E' and on E/E' *induced* by ∇ are the restriction of ∇ to E' and the connection which sends the class of a local holomorphic section s of E to the class of ∇s in $\Omega^1_X \otimes (E/E')$, respectively. The holomorphic connections on $\det E$ and on $\operatorname{Sym}^k E$ for $k \geq 1$ induced by ∇ are defined by the Leibniz rule on products of local holomorphic sections of E. For a holomorphic map $g : X' \to X$, the *pullback* $g^*\nabla$ is the holomorphic connection on g^*E with $(g^*\nabla)(g^*s) = g^*(\nabla s)$ for any local holomorphic section s of E.

Definition 2.2. Let X and F be complex manifolds, and let $n \geq 1$ be an integer. A *holomorphic fiber bundle* over X with fiber F is a holomorphic map $p : P \to X$ such that any point of X has an open neighborhood W with a biholomorphism $p^{-1}(W) \cong W \times F$ over W. A *holomorphic $\mathbb{P}^n$-bundle* over X is a holomorphic fiber bundle over X with fiber $\mathbb{P}^n$.

We say that a holomorphic $\mathbb{P}^n$-bundle $p : P \to X$ admits a *flat holomorphic connection* if there exist an open cover $\{W_i\}$ of X and biholomorphisms $p^{-1}(W_i) \cong W_i \times \mathbb{P}^n$ over W_i such that any transition map from $W_i \cap W_j$ to $\operatorname{Aut}(\mathbb{P}^n)$ is locally constant. For a holomorphic vector bundle E of rank $n+1$ on X, we denote by $\mathbb{P}(E) \to X$ the holomorphic $\mathbb{P}^n$-bundle of lines in E.

Equivalently, the holomorphic principal $\mathrm{PGL}_{n+1}(\mathbb{C})$-bundle of isomorphisms from $\mathbb{P}^n$ to the fibers of p carries a holomorphic connection with zero curvature (cf. [KN63, Chapter II, Theorem 9.1]). In this paper we only use the description by locally constant transition maps.

Lemma 2.3. *Let X be a simply connected complex manifold.*

(1) If a holomorphic vector bundle E on X admits a flat holomorphic connection ∇, then the evaluation map

$$\{s \in H^0(X, E) \mid \nabla s = 0\} \otimes_{\mathbb{C}} \mathcal{O}_X \to E$$

is an isomorphism. In particular, $E \cong \mathcal{O}_X^{\oplus \operatorname{rank} E}$.

(2) If a holomorphic $\mathbb{P}^n$-bundle $P \to X$ admits a flat holomorphic connection, then $P \cong X \times \mathbb{P}^n$ over X.

Proof. (1) Let $\mathbb{V} \subset E$ be the sheaf of local holomorphic sections s of E with $\nabla s = 0$. Since $R_\nabla = 0$, $\mathbb{V}$ is a local system of complex vector spaces of rank $\operatorname{rank} E$, and the natural map $\mathbb{V} \otimes_{\mathbb{C}} \mathcal{O}_X \to E$ is an isomorphism [Del70, Chapter I, Théorème 2.17]. Since X is simply connected, $\mathbb{V}$ is a constant sheaf, so the restriction map $H^0(X, \mathbb{V}) \to \mathbb{V}_x$ is an isomorphism for any $x \in X$. Since $H^0(X, \mathbb{V}) = \{s \in H^0(X, E) \mid \nabla s = 0\}$, the evaluation map is an isomorphism. Choosing a basis of $H^0(X, \mathbb{V})$, we obtain $E \cong \mathcal{O}_X^{\oplus \operatorname{rank} E}$.

(2) Let $p : P \to X$ be the projection. By Definition 2.2, there exist an open cover $\{W_i\}$ of X and biholomorphisms $\phi_i : p^{-1}(W_i) \cong W_i \times \mathbb{P}^n$ over W_i whose transition maps $g_{ij} : W_i \cap W_j \to \operatorname{Aut}(\mathbb{P}^n)$ are locally constant, where $\phi_i \circ \phi_j^{-1}(x, y) = (x, g_{ij}(x)y)$. Then (g_{ij}) is a Čech 1-cocycle with values in the constant sheaf of groups $\operatorname{Aut}(\mathbb{P}^n)$, and the classes of such cocycles correspond to

the conjugacy classes of homomorphisms $\pi_1(X) \to \mathrm{Aut}(\mathbb{P}^n)$, the monodromy representations. Since $\pi_1(X)$ is trivial, the class of (g_{ij}) is trivial (cf. [KN63, Chapter II, Corollary 9.2]): possibly refining $\{W_i\}$, there exist locally constant maps $h_i : W_i \to \mathrm{Aut}(\mathbb{P}^n)$ with $g_{ij} = h_i h_j^{-1}$ on $W_i \cap W_j$. Thus the biholomorphisms $(x, y) \to (x, h_i(x)^{-1}y)$ composed with ϕ_i glue to a biholomorphism $P \cong X \times \mathbb{P}^n$ over X. □

2.2. **An indecomposable bundle on a curve.**

Definition 2.4. Let X be a complex manifold, and let N, Q, and G be holomorphic vector bundles on X. We say that G is an *extension of Q by N* if there exists an exact sequence $0 \to N \to G \to Q \to 0$ of holomorphic vector bundles. We say that G is a *non-split extension of Q by N* if there exists such an exact sequence whose class in $\mathrm{Ext}^1(Q, N)$ is non-zero.

If X is compact, then G is a non-split extension of Q by N if and only if G is an extension of Q by N with $G \ncong N \oplus Q$, so in Theorem 1.4(4) and Lemma 2.5, where the bundles are on smooth projective curves, a non-split extension of Q by N may also be read as an extension of Q by N not isomorphic to $N \oplus Q$. Indeed, an exact sequence

$$0 \to N \to G \xrightarrow{p} Q \to 0$$

with class zero splits, so that $G \cong N \oplus Q$, and conversely, if $G \cong N \oplus Q$, then $\dim \mathrm{Hom}(Q, G)$ is the sum of $\dim \mathrm{Hom}(Q, N)$ and $\dim \mathrm{Hom}(Q, Q)$, which are finite since X is compact [Huy05, Corollary 4.1.14], so the last map p_* of the left exact sequence

$$0 \to \mathrm{Hom}(Q, N) \to \mathrm{Hom}(Q, G) \xrightarrow{p_*} \mathrm{Hom}(Q, Q)$$

is surjective, and a preimage of id_Q under p_* splits any such exact sequence.

The flat coefficients in Construction-Proposition 4.1 are built from the bundle E_0 of Lemma 2.5 below. The bundle E_0 carries a flat holomorphic connection, but it is not semistable (cf. [Bis95, Remark 3.7(iii)]), and the count in Section 4 depends on the instability of E_0.

Lemma 2.5. *Let C be a smooth projective curve of genus $g \geq 2$. Then there exist a line bundle $\mathcal{L}$ on C with $\mathcal{L}^{\otimes 2} \cong \omega_C$ and a holomorphic vector bundle E_0 on C which is a non-split extension of $\mathcal{L}^{-1}$ by $\mathcal{L}$. For any such $\mathcal{L}$ and E_0, the vector bundle E_0 is indecomposable, and E_0 admits a holomorphic connection which induces the trivial connection on $\det E_0 \cong \mathcal{O}_C$. Moreover, any holomorphic connection on E_0 is flat.*

Proof. By the exponential sequence, $\mathrm{Pic}^0(C) \cong H^1(C, \mathcal{O}_C)/H^1(C, \mathbb{Z})$ is a complex torus, so multiplication by 2 on $\mathrm{Pic}^0(C)$ is surjective. Thus for a line bundle $\mathcal{L}_0$ of degree $g-1$ there exists $\mathcal{M} \in \mathrm{Pic}^0(C)$ with $\mathcal{M}^{\otimes 2} \cong \omega_C \otimes \mathcal{L}_0^{-2}$, and $\mathcal{L} := \mathcal{L}_0 \otimes \mathcal{M}$ satisfies $\mathcal{L}^{\otimes 2} \cong \omega_C$. By [Har77, III, Propositions 6.7 and 6.3(c)] and Serre duality [Har77, III, Corollary 7.7 and Remark 7.12.1],

$$\mathrm{Ext}^1(\mathcal{L}^{-1}, \mathcal{L}) \cong \mathrm{Ext}^1(\mathcal{O}_C, \mathcal{L}^{\otimes 2}) \cong H^1(C, \omega_C) \cong H^0(C, \mathcal{O}_C)^\vee \cong \mathbb{C}$$

(the algebraic and analytic cohomology groups are isomorphic by GAGA [Ser56, n° 12, Théorème 1]), so a non-split extension E_0 of $\mathcal{L}^{-1}$ by $\mathcal{L}$ exists [Har77, III, Exercise 6.1]. Now fix such $\mathcal{L}$ and E_0, with an exact sequence

$$0 \to \mathcal{L} \xrightarrow{i} E_0 \xrightarrow{q} \mathcal{L}^{-1} \to 0$$

whose class is non-zero, and let h be a global endomorphism of E_0. Since $\deg \omega_C^{-1} = 2 - 2g < 0$, we have $\mathrm{Hom}(\mathcal{L}, \mathcal{L}^{-1}) = H^0(C, \omega_C^{-1}) = 0$, hence $q \circ h \circ i = 0$, so h preserves $\mathcal{L}$ and induces multiplication by some $\lambda \in \mathbb{C}$ on $\mathcal{L}$ and by some $\mu \in \mathbb{C}$ on $\mathcal{L}^{-1}$, since $H^0(C, \mathcal{O}_C) = \mathbb{C}$. If $\lambda \neq \mu$, then $q \circ (h - \mu \,\mathrm{id}_{E_0}) = 0$, so $h - \mu \,\mathrm{id}_{E_0} = i \circ r$ for some $r : E_0 \to \mathcal{L}$ with $r \circ i = (\lambda - \mu)\,\mathrm{id}_{\mathcal{L}}$, and $(\lambda - \mu)^{-1} r$ splits the exact sequence, a contradiction. Thus $\lambda = \mu$, and $h - \lambda \,\mathrm{id}_{E_0}$ vanishes on $\mathcal{L}$ and has image in $\mathcal{L}$, so $h = \lambda \,\mathrm{id}_{E_0} + N_s$, where $N_s := i \circ s \circ q$ for some $s \in \mathrm{Hom}(\mathcal{L}^{-1}, \mathcal{L}) = H^0(C, \omega_C)$. Since $q \circ i = 0$, we have $N_s N_{s'} = 0$ for any $s, s' \in H^0(C, \omega_C)$. If h is idempotent, then $h^2 = \lambda^2 \,\mathrm{id}_{E_0} + 2\lambda N_s = h$ implies $\lambda^2 = \lambda$ and $(2\lambda - 1)N_s = 0$, hence $\lambda \in \{0, 1\}$, $N_s = 0$, and $h \in \{0, \mathrm{id}_{E_0}\}$. A decomposition $E_0 \cong E_1 \oplus E_2$ with $E_1 \neq 0 \neq E_2$ would induce a

projection onto E_1, an idempotent different from 0 and id_{E_0}, so E_0 is indecomposable. Since $\det E_0 \cong \mathcal{L} \otimes \mathcal{L}^{-1} \cong \mathcal{O}_C$, we have $\deg E_0 = 0$, so E_0 admits a holomorphic connection ∇' by the Atiyah–Weil criterion [Wei38], [Ati57, Theorems 2, 8, and 10], see [HL19, Theorem 3.1] for the statement on any compact Riemann surface. The connection on $\det E_0 \cong \mathcal{O}_C$ induced by ∇' is $d + \eta$ for some $\eta \in H^0(C, \omega_C)$, and since $\operatorname{rank} E_0 = 2$, the holomorphic connection $\nabla_0 := \nabla' - \frac{1}{2}\eta \otimes \mathrm{id}_{E_0}$ induces the trivial connection d on $\det E_0$. Finally, the curvature of any holomorphic connection on E_0 is a section of $\Omega_C^2 \otimes \operatorname{End} E_0$, which is zero since $\Omega_C^2 = 0$. □

3. An elliptic surface over a curve of genus two

In this section, we construct the elliptic surface S and prove the properties of S used in Sections 4 and 5. The invariants ω_S, $R^1 f_* \mathcal{O}_S$, $e(S)$, and $\pi_1(S)$ of S and of the fibration $f : S \to C$ (Construction 3.2) enter the count of Section 4. Section 5 uses two further properties: loops in the fibers of f are null-homotopic in S, so that the fibers lift isomorphically to the universal cover of S, and holomorphic connections on bundles pulled back from C come from C.

3.1. The surface.

Lemma 3.1. *Let C be a smooth projective curve of genus two, let $\mathcal{A}$ be a line bundle of degree five on C, and let $\Sigma \subset \mathbb{P}^1$ be a finite subset. Then there exist $a_0, a_1 \in H^0(C, \mathcal{A})$ without common zeros such that the morphism $h : C \to \mathbb{P}^1$ defined by a_0, a_1 has no branch value in Σ.*

Proof. Since C has genus two and $\deg \mathcal{A} = 5 = 2 \cdot 2 + 1$, $\mathcal{A}$ is very ample [Har77, IV, Corollary 3.2], and in particular globally generated. Pick $a_0 \in H^0(C, \mathcal{A}) \backslash \{0\}$, and let $c_1, \dots, c_k$ be the zeros of a_0. For $1 \le i \le k$, the subspace $V_i \subset H^0(C, \mathcal{A})$ of sections vanishing at c_i is a proper linear subspace because $\mathcal{A}$ is globally generated. Pick $a_1 \in H^0(C, \mathcal{A}) \backslash \bigcup_{i=1}^k V_i$. Then a_0 and a_1 have no common zeros, so they define a morphism $h' : C \to \mathbb{P}^1$ with $h'^* \mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{A}$ and $\deg h' = 5$. Let $\Sigma' \subset \mathbb{P}^1$ be the set of branch values of h', which is finite because h' is non-constant.

Let $g \in \mathrm{PGL}_2(\mathbb{C})$ be represented by an invertible matrix $(g_{kl})_{0 \le k,l \le 1}$. Then $g \circ h'$ is defined by the sections $g_{00}a_0 + g_{01}a_1$ and $g_{10}a_0 + g_{11}a_1$ of $\mathcal{A}$, which have no common zeros since (g_{kl}) is invertible, and the set of branch values of $g \circ h'$ is $g(\Sigma')$. For $b \in \Sigma'$ and $\sigma \in \Sigma$, the set $Z_{b,\sigma} := \{g \in \mathrm{PGL}_2(\mathbb{C}) \mid g(b) = \sigma\}$ is a closed subset of $\mathrm{PGL}_2(\mathbb{C})$, and $Z_{b,\sigma} \ne \mathrm{PGL}_2(\mathbb{C})$ since $\mathrm{PGL}_2(\mathbb{C})$ acts transitively on $\mathbb{P}^1$. Since $\mathrm{PGL}_2(\mathbb{C})$ is an open subset of $\mathbb{P}^3$, it is irreducible, so the finite union of the proper closed subsets $Z_{b,\sigma}$ is not all of $\mathrm{PGL}_2(\mathbb{C})$. Pick

$$g \in \mathrm{PGL}_2(\mathbb{C}) \backslash \bigcup_{b \in \Sigma', \sigma \in \Sigma} Z_{b,\sigma}.$$

Then the sections $g_{00}a_0 + g_{01}a_1$ and $g_{10}a_0 + g_{11}a_1$ satisfy the lemma. □

Construction 3.2. Let C be a smooth projective curve of genus two and $\mathcal{A}$ a line bundle of degree five on C. Let $h : C \to \mathbb{P}^1$ be a morphism defined by two sections $a_0, a_1 \in H^0(C, \mathcal{A})$ without common zeros, so that $h^* \mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{A}$, such that no branch value of h is one of the four points $\infty, -3, -3\epsilon, -3\epsilon^2$, where ϵ is a primitive cube root of unity and a point $(t_0 : t_1) \in \mathbb{P}^1$ with $t_0 \ne 0$ is identified with t_1/t_0. Such a morphism h exists by Lemma 3.1. Let $(x : y : z)$ be the homogeneous coordinates on $\mathbb{P}^2$, let

$$S_0 := \{t_0(x^3 + y^3 + z^3) + t_1 xyz = 0\} \subset \mathbb{P}^2 \times \mathbb{P}^1$$

be the *Hesse pencil*, and let $f_0 : S_0 \to \mathbb{P}^1$ be the projection. Let

$$S := S_0 \times_{\mathbb{P}^1} C = \{a_0(x^3 + y^3 + z^3) + a_1 xyz = 0\} \subset \mathbb{P}^2 \times C,$$

let $f : S \to C$ be the projection, let $F_c := f^{-1}(c)$ for $c \in C$, and let $s : C \to S$ be the section of f given by the point $(1 : -1 : 0) \in \mathbb{P}^2$. Finally, let $\widetilde{C} \to C$ be the universal cover, let $\widetilde{S} := S \times_C \widetilde{C}$, and let $q : \widetilde{S} \to S$ be the projection.

The genus two is the smallest genus allowed by Lemma 2.5. The degree five makes $\mathcal{A}$ very ample, which is used in Lemmas 3.1 and 3.4. The condition on the branch values of h is used for the smoothness of S along the singular fibers of f in Lemma 3.3(1).

The section s is used in the proof of Proposition 3.6. We take the Hesse pencil because its singular members, its nine base points, and the smoothness of S can be checked directly from the equations. Only the non-constancy of the j-invariant is quoted from [AD09, §2].

Lemma 3.3. *Notation as in Construction 3.2. Then the following hold.*

(1) S is a smooth connected projective surface, and S is a member of the linear system $|\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A}|$ on $\mathbb{P}^2 \times C$.
(2) Any fiber F_c is a plane cubic curve which is either smooth or the union of three lines in general position, exactly twenty fibers of f are singular, and the j-invariant of the smooth fibers of f is a non-constant function on the open subset of C over which f is smooth.
(3) The topological Euler characteristic of S is 60.

Proof. Let $G_0 := x^3 + y^3 + z^3$, let $G_1 := xyz$, and let $\Sigma_0 := \{\infty, -3, -3\epsilon, -3\epsilon^2\} \subset \mathbb{P}^1$. For $t \in \mathbb{P}^1 \backslash \{\infty\}$ let $D_t := \{G_0 + tG_1 = 0\} \subset \mathbb{P}^2$, and let $D_\infty := \{G_1 = 0\}$. Then $f_0^{-1}(t) = D_t \times \{t\}$ for any $t \in \mathbb{P}^1$.

We first describe the curves D_t. The curve D_∞ is the union of the three coordinate lines, which have no common point. A direct calculation shows that

$$P(x, y, z) := x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \epsilon y + \epsilon^2 z)(x + \epsilon^2 y + \epsilon z).$$

The three linear factors are linearly independent, since their coefficient matrix is a Vandermonde matrix in $1, \epsilon, \epsilon^2$. For $k \in \{0, 1, 2\}$ we have $G_0 + (-3\epsilon^k)G_1 = P(x, \epsilon^k y, z)$, so D_t is the union of three lines in general position for any $t \in \Sigma_0$. Now let $t \in \mathbb{P}^1 \backslash \Sigma_0$, and suppose that $(x : y : z)$ is a singular point of D_t. Then

$$3x^2 + tyz = 3y^2 + txz = 3z^2 + txy = 0.$$

Multiplying the three left-hand sides by x, y, and z, respectively, we obtain

$$3x^3 = 3y^3 = 3z^3 = -txyz.$$

If $xyz = 0$, then $x = y = z = 0$, which is impossible. Thus, after scaling, $x = 1$ and $y^3 = z^3 = 1$, and $3 + tyz = 0$ implies $t^3 = -27$, so $t \in \Sigma_0$, a contradiction. Hence D_t is a smooth plane cubic for any $t \in \mathbb{P}^1 \backslash \Sigma_0$.

Next we show that S_0 is smooth. The base locus $\{G_0 = G_1 = 0\}$ consists of the nine points with exactly one zero coordinate and the other two coordinates u, v satisfying $u^3 + v^3 = 0$. Let $\Phi := t_0 G_0 + t_1 G_1$, so that $S_0 = \{\Phi = 0\}$. At a point $(p, t) \in S_0$ such that p is not a base point, the derivative of Φ in the $\mathbb{P}^1$-direction is non-zero: in the chart $t_0 = 1$ it is $G_1(p)$, and $G_1(p) = 0$ would imply $G_0(p) = -tG_1(p) = 0$, and in the chart $t_1 = 1$ it is $G_0(p)$, and $G_0(p) = 0$ would imply $G_1(p) = -t_0 G_0(p) = 0$. At a base point p, the derivative of Φ in the $\mathbb{P}^2$-directions at $(p, (t_0 : t_1))$ is $t_0 dG_0(p) + t_1 dG_1(p)$, which is non-zero since $dG_0(p)$ and $dG_1(p)$ are linearly independent: for $p = (1 : -\zeta : 0)$ with $\zeta^3 = 1$,

$$\nabla G_0(p) = (3, 3\zeta^2, 0), \quad \nabla G_1(p) = (0, 0, -\zeta),$$

and the other base points are obtained by permuting the coordinates. Thus S_0 is smooth. Since $T_{(p,t)} S_0 = \ker d\Phi_{(p,t)}$, the projection f_0 is a submersion at (p, t) if and only if $d\Phi_{(p,t)}$ does not vanish on $T_p \mathbb{P}^2 \times \{0\}$, that is, if and only if D_t is smooth at p. Hence f_0 is a submersion over $\mathbb{P}^1 \backslash \Sigma_0$.

(1) Let $C^\circ := h^{-1}(\mathbb{P}^1 \backslash \Sigma_0)$. Since f_0 is a submersion over $\mathbb{P}^1 \backslash \Sigma_0$, the base change $f^{-1}(C^\circ) \to C^\circ$ of f_0 is a submersion, and $f^{-1}(C^\circ)$ is smooth. Let $c \in C \backslash C^\circ$. Since $h(c)$ is not a branch value of h, h is a local isomorphism near c, so a neighborhood of F_c in S is isomorphic to a neighborhood of $f_0^{-1}(h(c))$ in S_0, which is smooth. Thus S is smooth. Since S is closed in $\mathbb{P}^2 \times C$, S is projective. Since $h^* t_k = a_k$ for $k = 0, 1$ under $h^* \mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{A}$, $S = (\mathrm{id}_{\mathbb{P}^2} \times h)^{-1}(S_0)$ is the zero

scheme of the section $a_0G_0 + a_1G_1$ of $\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A}$, so $S \in |\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A}|$. Finally, $F_c \cong D_{h(c)}$ is connected for any $c \in C$, since any two curves in $\mathbb{P}^2$ meet [Har77, I, Theorem 7.2]. If $S = U_1 \sqcup U_2$ with U_1, U_2 open and closed, then any fiber of f is contained in U_1 or in U_2, and since f is proper, $C = f(U_1) \sqcup f(U_2)$ with $f(U_1)$ and $f(U_2)$ closed. Since C is connected, U_1 or U_2 is empty. Thus S is connected.

(2) For any $c \in C$, $F_c \cong D_{h(c)}$ is a plane cubic curve which is either smooth or the union of three lines in general position, and F_c is singular if and only if $h(c) \in \Sigma_0$. Since no point of Σ_0 is a branch value of h and $\deg h = 5$, any point of Σ_0 has exactly five preimages under h, so exactly $4 \cdot 5 = 20$ fibers of f are singular, and C° is the open subset of C over which f is smooth. For $t \in \mathbb{P}^1 \backslash \Sigma_0$, let $j_0(t)$ be the j-invariant of D_t. The function j_0 is non-constant on $\mathbb{P}^1 \backslash \Sigma_0$ [AD09, §2]. Since the j-invariant of F_c is $j_0(h(c))$ for $c \in C^\circ$ and $h(C^\circ) = \mathbb{P}^1 \backslash \Sigma_0$, the j-invariant of the smooth fibers of f is non-constant on C°.

(3) Let F be a smooth fiber of f. By [BHPV04, Proposition III.11.4],

$$e(S) = e(C)e(F) + \sum_{c \in C \backslash C^\circ} \big(e(F_c) - e(F)\big).$$

Since F is an elliptic curve, $e(F) = 0$. For $c \in C \backslash C^\circ$, F_c is the union of three copies of $\mathbb{P}^1$ meeting pairwise in three distinct points, so $e(F_c) = 3 \cdot 2 - 3 = 3$. By (2), $e(S) = 20 \cdot 3 = 60$. □

Lemma 3.4. *Notation as in Construction 3.2. Then $f_* : \pi_1(S) \to \pi_1(C)$ is an isomorphism, $q : \widetilde{S} \to S$ is a universal cover of S, and q maps any fiber of the projection $\widetilde{S} \to \widetilde{C}$ isomorphically onto a fiber of f.*

Proof. The line bundle $\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A}$ on $\mathbb{P}^2 \times C$ is very ample: it is the pullback of $\mathcal{O}(1)$ under the composition of the product of the 3-uple embedding of $\mathbb{P}^2$ and the embedding of C defined by $\mathcal{A}$ [Har77, IV, Corollary 3.2] with the Segre embedding [Har77, II, Exercises 5.11 and 5.13]. For an ample divisor $W \geq 0$ on a smooth projective threefold X, the inclusion $W \to X$ induces an isomorphism $\pi_1(W) \to \pi_1(X)$ by the Lefschetz theorem for homotopy groups [Laz04, Theorem 3.1.21]. By Lemma 3.3(1), S is a member of the very ample linear system $|\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A}|$, so S is an ample divisor on $\mathbb{P}^2 \times C$, and the inclusion $S \to \mathbb{P}^2 \times C$ induces an isomorphism $\pi_1(S) \to \pi_1(\mathbb{P}^2 \times C)$. Since $\mathbb{P}^2$ is simply connected, the projection $\mathbb{P}^2 \times C \to C$ induces an isomorphism $\pi_1(\mathbb{P}^2 \times C) \to \pi_1(C)$. The composition of the inclusion and the projection is f, so f_* is an isomorphism.

The projection q is a covering map, as the pullback of the covering map $\widetilde{C} \to C$. Let $x \in S$, and let γ be a loop in S based at x. A lift of γ to $\widetilde{S}$ starting at $(x, \tilde{c})$ is the path $(\gamma, \tilde{\gamma})$, where $\tilde{\gamma}$ is the lift of $f \circ \gamma$ to $\widetilde{C}$ starting at $\tilde{c}$. Thus the lift of γ is closed if and only if $f_*[\gamma] = 1$, that is, if and only if $[\gamma] = 1$. Hence $q_*\pi_1(\widetilde{S}, (x, \tilde{c})) = 1$, and since q_* is injective, any connected component of $\widetilde{S}$ is simply connected. Let $\tilde{c}_1, \tilde{c}_2 \in \widetilde{C}$ be two points over $f(x)$, and let δ be the image in C of a path from $\tilde{c}_1$ to $\tilde{c}_2$ in $\widetilde{C}$. Since f_* is surjective, δ is homotopic to $f \circ \gamma$ for a loop γ in S based at x, and then the lift of γ starting at $(x, \tilde{c}_1)$ ends at $(x, \tilde{c}_2)$. Hence the connected component of $\widetilde{S}$ containing $(x, \tilde{c}_1)$ contains $(x, \tilde{c}_2)$, so $q^{-1}(x)$ is contained in one connected component of $\widetilde{S}$. Since S is connected, any connected component of $\widetilde{S}$ maps onto S, so $\widetilde{S}$ is connected. Thus q is a universal cover of S.

Finally, for $\tilde{c} \in \widetilde{C}$ with image $c \in C$, the fiber of $\widetilde{S} \to \widetilde{C}$ over $\tilde{c}$ is $F_c \times \{\tilde{c}\}$, and q maps it isomorphically onto F_c. □

Lemma 3.5. *Notation as in Construction 3.2. Then $f_*\mathcal{O}_S = \mathcal{O}_C$, $R^1 f_* \mathcal{O}_S \cong \mathcal{A}^{-1}$, and $\omega_S \cong f^*(\omega_C \otimes \mathcal{A})$.*

Proof. Let $p : \mathbb{P}^2 \times C \to C$ be the projection. By Lemma 3.3(1), there is an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-3) \boxtimes \mathcal{A}^{-1} \to \mathcal{O}_{\mathbb{P}^2 \times C} \to \mathcal{O}_S \to 0.$$

Since S is closed in $\mathbb{P}^2 \times C$ and f is the restriction of p, $R^i p_* \mathcal{O}_S = R^i f_* \mathcal{O}_S$ for any i. For a line bundle $\mathcal{N}$ on C and an integer k, we have

$$R^i p_*(\mathcal{O}_{\mathbb{P}^2}(k) \boxtimes \mathcal{N}) \cong H^i(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k)) \otimes \mathcal{N}$$

by flat base change [Har77, III, Proposition 9.3] and the projection formula [Har77, III, Exercise 8.3]. By [Har77, III, Theorem 5.1], $H^i(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$ is $\mathbb{C}$ for $i = 0$ and 0 for $i > 0$, and $H^i(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3))$ is $\mathbb{C}$ for $i = 2$ and 0 for $i \neq 2$. Thus the long exact sequence of higher direct images under p reads

$$0 \to \mathcal{O}_C \to f_* \mathcal{O}_S \to 0 \to 0 \to R^1 f_* \mathcal{O}_S \to \mathcal{A}^{-1} \to 0.$$

Hence $f_* \mathcal{O}_S = \mathcal{O}_C$ and $R^1 f_* \mathcal{O}_S \cong \mathcal{A}^{-1}$. Since $\omega_{\mathbb{P}^2 \times C} \cong \mathcal{O}_{\mathbb{P}^2}(-3) \boxtimes \omega_C$ [Har77, II, Exercise 8.3(b) and Example 8.20.1], the adjunction formula [Har77, II, Proposition 8.20] implies

$$\omega_S \cong \big(\omega_{\mathbb{P}^2 \times C} \otimes (\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{A})\big)|_S \cong (\mathcal{O}_{\mathbb{P}^2} \boxtimes (\omega_C \otimes \mathcal{A}))|_S = f^*(\omega_C \otimes \mathcal{A}).$$

□

3.2. **Holomorphic one-forms and descent of connections.** In this subsection, we show that holomorphic connections on bundles pulled back from C descend to C. Section 5 descends to C a holomorphic connection on a bundle pulled back from C, and for the descent we need that, for any open subset $W \subset C$ and any holomorphic vector bundle N on W, any holomorphic f^*N-valued 1-form on $f^{-1}(W)$ is $f^*\lambda$ for an N-valued holomorphic 1-form λ on W (Proposition 3.6). The non-constancy of the j-invariant in Lemma 3.3(2) is used only here: on a product $C \times F$ with an elliptic curve $F = \mathbb{C}/\Lambda$, the 1-form $\mathrm{pr}_2^* dz$ is not the pullback of a 1-form on C.

Proposition 3.6. *Notation as in Construction 3.2. Then the pullback of holomorphic 1-forms induces an isomorphism $\omega_C \to f_* \Omega^1_S$. In particular, for any open subset $W \subset C$ and any holomorphic vector bundle N on W, the pullback map*

$$H^0(W, \omega_C \otimes N) \to H^0(f^{-1}(W), \Omega^1_S \otimes f^*N)$$

is an isomorphism.

Proof. Let $C^\circ \subset C$ be the open subset over which f is smooth, and let $j : C^\circ \to \mathbb{C}$ be the j-invariant of the fibers of f. By Lemma 3.3(2), $C \backslash C^\circ$ consists of twenty points and j is non-constant, so the set $C_1 := \{c \in C^\circ \mid dj(c) \neq 0\}$ is the complement of a closed discrete subset of C°.

We first show that $\dim H^0(F_c, \Omega^1_S|_{F_c}) = 1$ for any $c \in C_1$, and that $H^0(F_c, \Omega^1_S|_{F_c})$ is spanned by $(df)^* v$ for a non-zero $v \in T^*_c C$. Since f is a submersion along F_c, there is an exact sequence

$$0 \to T_{F_c} \to T_S|_{F_c} \to T_c C \otimes \mathcal{O}_{F_c} \to 0,$$

and the Kodaira–Spencer map $\rho : T_c C \to H^1(F_c, T_{F_c})$ of f at c is the boundary map of the associated long exact sequence [Huy05, Definition 6.2.6]. We show that $\rho \neq 0$. By Ehresmann's theorem, f is a locally trivial fibration over C°, so on a connected neighborhood $W_0 \subset C^\circ$ of c there exist loops $\gamma_1(c'), \gamma_2(c')$ in $F_{c'}$ depending continuously on $c' \in W_0$ whose classes form a basis of $H_1(F_{c'}, \mathbb{Z})$. Let $\xi_{c'}$ be the holomorphic 1-form on $F_{c'}$ with $\int_{\gamma_1(c')} \xi_{c'} = 1$, which exists and is unique since $F_{c'}$ has genus one and the periods of a non-zero holomorphic 1-form on $F_{c'}$ are $\mathbb{R}$-linearly independent [GH78, Chapter 2, §2, p. 228], and let $\tau(c') := \int_{\gamma_2(c')} \xi_{c'}$. Under the isomorphisms $H^1(F_{c'}, \mathbb{C}) \cong \mathbb{C}^2$ given by integration over $\gamma_1(c')$ and $\gamma_2(c')$, the line $H^{1,0}(F_{c'})$ is spanned by $(1, \tau(c'))$ and depends holomorphically on $c' \in W_0$ [Voi02, Theorem 10.9], so τ is holomorphic on W_0. For any $c' \in W_0$ and a point $x_0 \in F_{c'}$, the map $x \to \int_{x_0}^x \xi_{c'}$ is an isomorphism $F_{c'} \to \mathbb{C}/(\mathbb{Z} + \tau(c')\mathbb{Z})$ by Abel's theorem and Jacobi inversion in genus one [GH78, Chapter 2, §2, p. 238], so $\tau(c') \notin \mathbb{R}$. Since τ is continuous and W_0 is connected, $\operatorname{Im} \tau$ has constant sign on W_0, so, possibly replacing γ_2 with $-\gamma_2$, we may assume that $\operatorname{Im} \tau > 0$. Then $j = J \circ \tau$ on W_0, where J is the holomorphic function on the upper half-plane whose value at w is the j-invariant of $\mathbb{C}/(\mathbb{Z} + w\mathbb{Z})$ [Har77, IV, §4, pp. 326–327], [Ser73, VII, §3.3, Proposition 5]. Hence

$dj(c) = J'(\tau(c))d\tau(c) \neq 0$ implies $d\tau(c) \neq 0$. Pick $u \in T_cC$ with $d\tau(c)(u) \neq 0$. Since $(0, d\tau(c)(u))$ is not a multiple of $(1, \tau(c))$, the derivative of the line $H^{1,0}(F_{c'})$ along u, which is a map

$$H^{1,0}(F_c) \to H^1(F_c, \mathbb{C})/H^{1,0}(F_c) \cong H^1(F_c, \mathcal{O}_{F_c}),$$

is non-zero. By [Voi02, Theorem 10.21], the derivative is the cup product with $\rho(u) \in H^1(F_c, T_{F_c})$ composed with the contraction $T_{F_c} \otimes \Omega^1_{F_c} \to \mathcal{O}_{F_c}$. Hence $\rho(u) \neq 0$.

Since $\rho \neq 0$, the sequence of tangent bundles above does not split, and so the dual sequence

$$0 \to T_c^*C \otimes \mathcal{O}_{F_c} \xrightarrow{(df)^*} \Omega^1_S|_{F_c} \to \Omega^1_{F_c} \to 0$$

does not split. Since F_c is an elliptic curve, $\Omega^1_{F_c} \cong \mathcal{O}_{F_c}$, so the extension class of the dual sequence is a non-zero element $e_c \in H^1(F_c, \mathcal{O}_{F_c}) \otimes T_c^*C$, and the connecting map

$$\partial : H^0(F_c, \Omega^1_{F_c}) \cong H^0(F_c, \mathcal{O}_{F_c}) \to H^1(F_c, \mathcal{O}_{F_c}) \otimes T_c^*C$$

sends 1 to $e_c \neq 0$. Thus ∂ is injective, and

$$H^0(F_c, \Omega^1_S|_{F_c}) = (df)^*\big(H^0(F_c, \mathcal{O}_{F_c}) \otimes T_c^*C\big) = (df)^*T_c^*C,$$

which is one-dimensional.

Now we prove that $\omega_C \to f_*\Omega^1_S$ is an isomorphism. Since $f \circ s = \mathrm{id}_C$, we have $s^*f^*\lambda = \lambda$ for any holomorphic 1-form λ on an open subset of C, so the pullback is injective. Let $W \subset C$ be an open subset, let θ be a holomorphic 1-form on $f^{-1}(W)$, and let $\lambda := s^*\theta$. Let $c \in W \cap C_1$. By the previous paragraph, there exists $v_c \in T_c^*C$ such that $\theta_x = (df_x)^*v_c$ for any $x \in F_c$. At $x = s(c)$, we have

$$\lambda(c) = (ds_c)^*\theta_{s(c)} = (ds_c)^*(df_{s(c)})^*v_c = (d(f \circ s)_c)^*v_c = v_c.$$

Thus $\theta = f^*\lambda$ on F_c, and so $\theta - f^*\lambda$ vanishes on $f^{-1}(W \cap C_1)$. We show that $f^{-1}(W \cap C_1)$ is dense in $f^{-1}(W)$. Let $U \subset f^{-1}(W)$ be a non-empty open subset. Since $f^{-1}(C \backslash C^\circ)$ is a union of twenty curves, there exists $x \in U$ with $f(x) \in C^\circ$. Since f is a submersion at x, $f(U)$ contains a non-empty open subset of $W \cap C^\circ$, which is not contained in the discrete set $C^\circ \backslash C_1$. Thus $U \cap f^{-1}(W \cap C_1) \neq \emptyset$. Hence $\theta = f^*\lambda$ on $f^{-1}(W)$, and $\omega_C \to f_*\Omega^1_S$ is surjective.

For the pullback map on $H^0(W, \omega_C \otimes N)$, injectivity follows from $s^*f^*\lambda = \lambda$, applied to the coefficients in a local trivialization of N. Let $\sigma \in H^0(f^{-1}(W), \Omega^1_S \otimes f^*N)$. Let $W' \subset W$ be an open subset over which N is trivial, and let $e_1, \dots, e_r \in H^0(W', N)$ form a basis of N at any point of W'. Then $\sigma = \sum_{i=1}^r \theta_i \otimes f^*e_i$ on $f^{-1}(W')$ for holomorphic 1-forms θ_i on $f^{-1}(W')$, and $\theta_i = f^*\lambda_i$ for holomorphic 1-forms λ_i on W'. Let

$$\mu_{W'} := \sum_{i=1}^r \lambda_i \otimes e_i \in H^0(W', \omega_C \otimes N),$$

so that $f^*\mu_{W'} = \sigma$ on $f^{-1}(W')$. For two such open subsets W' and W'', $f^*(\mu_{W'} - \mu_{W''}) = 0$ on $f^{-1}(W' \cap W'')$, so $\mu_{W'} = \mu_{W''}$ on $W' \cap W''$ by injectivity. Thus the sections $\mu_{W'}$ glue to a section of $\omega_C \otimes N$ over W whose pullback is σ. □

Corollary 3.7. *Notation as in Construction 3.2, and let N be a holomorphic vector bundle on C. Then for any holomorphic connection ∇' on f^*N, there exists a unique holomorphic connection ∇_N on N with $\nabla' = f^*\nabla_N$. In particular, any holomorphic connection on f^*N is flat.*

Proof. Let $W \subset C$ be an open subset over which N is trivial, and let $e_1, \dots, e_r \in H^0(W, N)$ form a basis of N at any point of W. Write

$$\nabla'(f^*e_k) = \sum_{i=1}^r \theta_{ik} \otimes f^*e_i, \quad 1 \leq k \leq r,$$

where θ_{ik} are holomorphic 1-forms on $f^{-1}(W)$. By Proposition 3.6, $\theta_{ik} = f^*\gamma_{ik}$ for unique holomorphic 1-forms γ_{ik} on W. Let $\Gamma := (\gamma_{ik})$, and let $\nabla_W := d + \Gamma$ be the holomorphic connection on $N|_W$ with $\nabla_W e_k = \sum_{i=1}^r \gamma_{ik} \otimes e_i$. By Definition 2.1, $(f^*\nabla_W)(f^*e_k) = f^*(\nabla_W e_k) = \nabla'(f^*e_k)$

for any k. Since a holomorphic connection is determined by the images of $f^*e_1, \dots, f^*e_r$ by the Leibniz rule, $f^*\nabla_W = \nabla'$ on $f^{-1}(W)$.

Let $W' \subset C$ be another open subset over which N is trivial, and let $\nabla_{W'}$ be the holomorphic connection on $N|_{W'}$ with $f^*\nabla_{W'} = \nabla'$ on $f^{-1}(W')$ given by the first paragraph of this proof applied to W'. Then $\nabla_W - \nabla_{W'} \in H^0(W \cap W', \omega_C \otimes \operatorname{End} N)$, and $f^*(\nabla_W - \nabla_{W'}) = \nabla' - \nabla' = 0$ on $f^{-1}(W \cap W')$. By Proposition 3.6 applied to $\operatorname{End} N$, $\nabla_W = \nabla_{W'}$ on $W \cap W'$. Thus the connections ∇_W glue to a holomorphic connection ∇_N on N with $f^*\nabla_N = \nabla'$. If ∇''_N is another holomorphic connection on N with $f^*\nabla''_N = \nabla'$, then $\nabla_N - \nabla''_N \in H^0(C, \omega_C \otimes \operatorname{End} N)$ has zero pullback to S, so $\nabla''_N = \nabla_N$ by Proposition 3.6.

Finally, $R_{\nabla_N} \in H^0(C, \Omega^2_C \otimes \operatorname{End} N) = 0$ since C is a curve. Let $W \subset C$ be an open subset over which N is trivial, and write $\nabla_N = d + \Gamma$ with respect to sections $e_1, \dots, e_r \in H^0(W, N)$ forming a basis of N at any point of W. Then $\nabla' = d + f^*\Gamma$ with respect to $f^*e_1, \dots, f^*e_r$, so, by a direct calculation from Definition 2.1, the curvature of ∇' on $f^{-1}(W)$ is

$$d(f^*\Gamma) + f^*\Gamma \wedge f^*\Gamma = f^*(d\Gamma + \Gamma \wedge \Gamma) = f^*R_{\nabla_N} = 0.$$

Thus ∇' is flat. □

4. An extension with a holomorphic connection

In this section, we construct E and ∇. The key input is a non-zero holomorphic 2-form with values in a flat vector bundle which is exact for the flat connection.

4.1. **Holomorphic two-forms and cohomology of a local system.**

In this subsection, we construct a flat vector bundle M on C and show that some non-zero holomorphic 2-form on S with values in f^*M is exact for the pullback connection.

Construction-Proposition 4.1. Let C be the curve of Construction 3.2, let $\mathcal{L}$ be a line bundle on C with $\mathcal{L}^{\otimes 2} \cong \omega_C$, let E_0 be a holomorphic vector bundle on C which is a non-split extension of $\mathcal{L}^{-1}$ by $\mathcal{L}$, and let ∇_0 be a holomorphic connection on E_0 which induces the trivial connection on $\det E_0 \cong \mathcal{O}_C$ (see Lemma 2.5). Let $M := \operatorname{Sym}^{260} E_0$, and let ∇_M be the holomorphic connection on M induced by ∇_0. Then M is a holomorphic vector bundle of rank 261, ∇_M is flat, and ∇_M induces the trivial connection on $\det M \cong \mathcal{O}_C$.

Proof. Let e_1, e_2 be a local basis of E_0 consisting of holomorphic sections such that $e_1 \wedge e_2$ corresponds to 1 under $\det E_0 \cong \mathcal{O}_C$, and write $\nabla_0 e_i = \sum_{k=1}^2 \gamma_{ki} \otimes e_k$. Since ∇_0 induces the trivial connection on $\det E_0$, we have $(\gamma_{11} + \gamma_{22}) \otimes (e_1 \wedge e_2) = 0$, so the connection matrix (γ_{ki}) has trace zero. The monomials $e_1^a e_2^{260-a}$, $0 \le a \le 260$, form a local basis of M, so $\operatorname{rank} M = 261$. By the Leibniz rule, the connection matrix of ∇_M in the basis $(e_1^a e_2^{260-a})_{0 \le a \le 260}$ is the image of (γ_{ki}) under the symmetric power representation $\mathfrak{gl}_2 \to \mathfrak{gl}_{261}$, which maps $\mathfrak{sl}_2$ into $\mathfrak{sl}_{261}$. Thus the connection matrix of ∇_M has trace zero, and ∇_M maps the local section $\sigma := \bigwedge_{a=0}^{260} e_1^a e_2^{260-a}$ of $\det M$ to 0. Two such local bases of E_0 differ by a holomorphic map g to $\mathrm{SL}_2(\mathbb{C})$, and

$$\det \operatorname{Sym}^{260} g = (\det g)^{260 \cdot 261/2} = 1.$$

Hence the local sections σ glue to a nowhere vanishing horizontal section of $\det M$. Therefore $\det M \cong (\det E_0)^{\otimes (260 \cdot 261/2)} \cong \mathcal{O}_C$, and ∇_M induces the trivial connection on $\det M$. Finally, ∇_M is flat since $R_{\nabla_M} \in H^0(C, \Omega^2_C \otimes \operatorname{End} M)$ and $\Omega^2_C = 0$. □

Proposition 4.4 compares the lower bound 17816 of Lemma 4.2 with the upper bound 17748 of Lemma 4.3 for local systems of rank 261. The lower bound is larger because E_0 is unstable, so that M has graded pieces of positive degree.

Lemma 4.2. *Notation as in Construction 3.2 and Construction-Proposition 4.1. Then*

$$h^0(S, \Omega^2_S \otimes f^*M) \ge 17816.$$

Proof. Let $m := 130$, so that $M = \operatorname{Sym}^{2m} E_0$. Since E_0 is an extension of $\mathcal{L}^{-1}$ by $\mathcal{L}$, M has holomorphic subbundles

$$0 = M_{-1} \subset M_0 \subset M_1 \subset \cdots \subset M_{2m} = M$$

with $M_j/M_{j-1} \cong \mathcal{L}^{\otimes(2m-j)} \otimes \mathcal{L}^{\otimes(-j)} = \mathcal{L}^{\otimes(2m-2j)}$ for $0 \le j \le 2m$ [Har77, II, Exercise 5.16(c)]. Since $\mathcal{L}^{\otimes 2} \cong \omega_C$, for $0 \le j \le m$ we have

$$(M_j/M_{j-1}) \otimes \omega_C \otimes \mathcal{A} \cong \omega_C^{\otimes(m-j+1)} \otimes \mathcal{A},$$

and

$$\deg\left(\omega_C^{\otimes(m-j+1)} \otimes \mathcal{A}\right) = 2(m-j+1) + 5 > 2 = 2g(C) - 2.$$

Thus $H^1(C, \omega_C^{\otimes(m-j+1)} \otimes \mathcal{A})$, which is dual to $H^0(C, \omega_C^{\otimes(j-m)} \otimes \mathcal{A}^{-1})$ by Serre duality [Har77, III, Corollary 7.7 and Remark 7.12.1], vanishes, and by the Riemann–Roch theorem [Har77, IV, Theorem 1.3],

$$h^0(C, \omega_C^{\otimes(m-j+1)} \otimes \mathcal{A}) = 2(m-j+1) + 5 + 1 - 2 = 2(m-j) + 6.$$

By induction on j, the long exact sequences of cohomology of

$$0 \to M_{j-1} \otimes \omega_C \otimes \mathcal{A} \to M_j \otimes \omega_C \otimes \mathcal{A} \to \omega_C^{\otimes(m-j+1)} \otimes \mathcal{A} \to 0$$

for $0 \le j \le m$ imply that $H^1(C, M_j \otimes \omega_C \otimes \mathcal{A}) = 0$ and that $h^0(C, M_j \otimes \omega_C \otimes \mathcal{A})$ is the sum of the dimensions above. In particular,

$$h^0(C, M_m \otimes \omega_C \otimes \mathcal{A}) = \sum_{j=0}^{m} \big(2(m-j) + 6\big) = (m+1)(m+6) = 131 \cdot 136 = 17816.$$

Since $M_m \otimes \omega_C \otimes \mathcal{A}$ is a subsheaf of $M \otimes \omega_C \otimes \mathcal{A}$, we have $h^0(C, M \otimes \omega_C \otimes \mathcal{A}) \ge 17816$. By Lemma 3.5, we have $f_*\mathcal{O}_S = \mathcal{O}_C$ and

$$\Omega_S^2 \otimes f^*M = \omega_S \otimes f^*M \cong f^*(M \otimes \omega_C \otimes \mathcal{A}),$$

so by the projection formula [Har77, III, Exercise 8.3] (by GAGA [Ser56, n° 12, Théorèmes 1–3], we may compute h^0 in the algebraic category),

$$h^0(S, \Omega_S^2 \otimes f^*M) = h^0(C, M \otimes \omega_C \otimes \mathcal{A}) \ge 17816.$$

□

Lemma 4.3. *Notation as in Construction 3.2. Let $\mathbb{W}$ be a local system of complex vector spaces of rank r on S. Then $\dim_{\mathbb{C}} H^2(S, \mathbb{W}) \le 68r$.*

Proof. For any i, let $b_i(\mathbb{W}) := \dim_{\mathbb{C}} H^i(S, \mathbb{W})$. By [Bre97, III, Theorem 1.1], $H^i(S, \mathbb{W})$ is the singular cohomology of S with local coefficients in $\mathbb{W}$, so we may compute it with cellular cochains (see [Hat02, §3.H] for cellular (co)homology with local coefficients). Fix a structure of finite CW complex on the compact real 4-manifold S with cells of dimension at most 4, for instance from a smooth triangulation. The cellular cochain complex of S with coefficients in $\mathbb{W}$ has an r-dimensional summand for each i-cell in degree i. Since the topological Euler characteristic of S is 60 (Lemma 3.3(3)), we obtain

$$\sum_{i=0}^{4} (-1)^i b_i(\mathbb{W}) = 60r.$$

We show that $b_1(\mathbb{W}) \le 4r$ and $\dim_{\mathbb{C}} H_1(S, \mathbb{W}) \le 4r$, where $H_1(S, \mathbb{W})$ is singular homology with local coefficients. Let Y be a CW complex which contains S as a subcomplex, such that the cells of $Y \backslash S$ are of dimension at least 3 and kill the higher homotopy groups of S, so that Y is a $K(\pi_1(S), 1)$ ([Hat02, Example 4.16] with $n = 1$). Since $\pi_1(S) \to \pi_1(Y)$ is an isomorphism, $\mathbb{W}$ extends to a local system $\mathbb{W}_Y$ on Y. Since the cells of $Y \backslash S$ are of dimension at least 3, the

2-skeleton of Y is the 2-skeleton of S, so the cellular cochain complexes and the cellular chain complexes of $(S, \mathbb{W})$ and of $(Y, \mathbb{W}_Y)$ coincide in degrees at most 2. Hence

$$H^1(S, \mathbb{W}) \cong H^1(Y, \mathbb{W}_Y) \quad \text{and} \quad H_1(S, \mathbb{W}) \cong H_1(Y, \mathbb{W}_Y).$$

The curve C is a $K(\pi_1(C), 1)$, because the universal cover of C is contractible. Since $\pi_k(C) = 0$ for $k \geq 2$ and the cells of $Y \backslash S$ are of dimension at least 3, f extends to a map $Y \to C$ [Hat02, Lemma 4.7], which induces the isomorphism $f_* : \pi_1(S) \to \pi_1(C)$ of Lemma 3.4 on fundamental groups. Hence $Y \to C$ is a homotopy equivalence by the Whitehead theorem [Hat02, Theorem 4.5], and $\mathbb{W}_Y$ is the pullback of a local system $\mathbb{W}_C$ of rank r on C. The curve C of genus two has a CW structure with one 0-cell, four 1-cells, and one 2-cell, so the spaces of cellular 1-cochains and 1-chains of C with coefficients in $\mathbb{W}_C$ are of dimension $4r$. Therefore

$$b_1(\mathbb{W}) = \dim_{\mathbb{C}} H^1(C, \mathbb{W}_C) \leq 4r \quad \text{and} \quad \dim_{\mathbb{C}} H_1(S, \mathbb{W}) = \dim_{\mathbb{C}} H_1(C, \mathbb{W}_C) \leq 4r.$$

Since S is a closed orientable 4-manifold, Poincaré duality with local coefficients [Bre97, V, 9.7] implies that $H^3(S, \mathbb{W}) \cong H_1(S, \mathbb{W})$. Here, for the locally constant sheaf $\mathbb{W}$, the homology of [Bre97] is the classical singular homology with local coefficients [Bre97, V, p. 356, after Corollary 12.17]. Thus $b_3(\mathbb{W}) \leq 4r$, and

$$b_2(\mathbb{W}) = 60r - b_0(\mathbb{W}) + b_1(\mathbb{W}) + b_3(\mathbb{W}) - b_4(\mathbb{W}) \leq 60r + 4r + 4r = 68r.$$

□

For a holomorphic vector bundle N on S, we write $A^k(N)$ for the sheaf of smooth N-valued k-forms on S.

Proposition 4.4. *Notation as in Construction 3.2 and Construction-Proposition 4.1. Then there exist a non-zero section $\varphi \in H^0(S, \Omega_S^2 \otimes f^*M)$ and a smooth f^*M-valued 1-form η on S such that $d_{f^*\nabla_M}\eta = \varphi$.*

Proof. Since ∇_M is flat (Construction-Proposition 4.1), so is $f^*\nabla_M$. Let $\mathbb{W}$ be the sheaf of local holomorphic sections s of f^*M with $(f^*\nabla_M)s = 0$. Then $\mathbb{W}$ is a local system of rank 261 on S, and the natural map $\mathbb{W} \otimes_{\mathbb{C}} \mathcal{O}_S \to f^*M$ is an isomorphism [Del70, Chapter I, Théorème 2.17].

In a local basis of f^*M consisting of sections of $\mathbb{W}$, $d_{f^*\nabla_M}$ is the componentwise exterior derivative d. Thus, by the Poincaré lemma, the complex $(A^\bullet(f^*M), d_{f^*\nabla_M})$ is a fine resolution of $\mathbb{W}$, and the cohomology of the complex $\Gamma(S, A^\bullet(f^*M))$ of global sections is the sheaf cohomology $H^*(S, \mathbb{W})$ [Bre97, III, §3].

Let $\psi \in H^0(S, \Omega_S^2 \otimes f^*M)$. Then $\bar\partial_{f^*M}\psi = 0$ since ψ is holomorphic, and $(f^*\nabla_M)\psi$ is an f^*M-valued form of type $(3,0)$ on the surface S, hence zero. Thus $d_{f^*\nabla_M}\psi = 0$, and ψ defines a class $[\psi] \in H^2(S, \mathbb{W})$. By Lemma 4.3 with $r = 261$ and by Lemma 4.2,

$$\dim_{\mathbb{C}} H^2(S, \mathbb{W}) \leq 68 \cdot 261 = 17748 < 17816 \leq h^0(S, \Omega_S^2 \otimes f^*M).$$

Hence the linear map $H^0(S, \Omega_S^2 \otimes f^*M) \to H^2(S, \mathbb{W})$, $\psi \to [\psi]$, has a non-zero kernel. Let φ be a non-zero element of the kernel. Then $[\varphi] = 0$, so there exists a global section η of $A^1(f^*M)$ with $d_{f^*\nabla_M}\eta = \varphi$. □

4.2. **The extension.** In this subsection, we construct E and ∇ from the forms of Proposition 4.4. Later sections use only the conclusions of Construction-Proposition 4.5, not the specific choice of φ and η.

Construction-Proposition 4.5. Notation as in Construction-Proposition 4.1 and Construction 3.2. Let φ be a non-zero element of $H^0(S, \Omega_S^2 \otimes f^*M)$, and let η be a smooth f^*M-valued 1-form on S with $d_{f^*\nabla_M}\eta = \varphi$ (see Proposition 4.4). Write $\eta = \alpha + \beta$, where α is of type $(1,0)$ and β is of type $(0,1)$, and let E be the smooth complex vector bundle $f^*M \oplus \mathcal{O}_S$ with the $(0,1)$-operator

$$\bar\partial_E := \begin{pmatrix} \bar\partial_{f^*M} & \beta \\ 0 & \bar\partial \end{pmatrix}.$$

Then the following hold.

(1) $\bar\partial_E$ defines a structure of holomorphic vector bundle of rank 262 on E, the inclusion of f^*M and the projection to $\mathcal{O}_S$ are holomorphic, and the class of the extension $0 \to f^*M \to E \to \mathcal{O}_S \to 0$ in $H^1(S, f^*M)$ is represented by β.
(2) There exists a unique holomorphic connection ∇ on E with

$$d_\nabla = \begin{pmatrix} d_{f^*\nabla_M} & \eta \\ 0 & d \end{pmatrix}.$$

Moreover, ∇ preserves f^*M, induces $f^*\nabla_M$ on f^*M and d on $\mathcal{O}_S$, and

$$R_\nabla = \begin{pmatrix} 0 & \varphi \\ 0 & 0 \end{pmatrix} \neq 0.$$

Proof. We write a smooth E-valued form as (u, g), where u is an f^*M-valued form and g is a form, and we let the matrix operators act by

$$\bar\partial_E(u, g) = (\bar\partial_{f^*M} u + \beta \wedge g, \bar\partial g) \quad \text{and} \quad d_\eta(u, g) := (d_{f^*\nabla_M} u + \eta \wedge g, dg),$$

where d_η denotes the operator in (2). We also write $f^*\nabla_M$ for the $(1,0)$ part of $d_{f^*\nabla_M}$ on smooth forms, so that $d_{f^*\nabla_M} = f^*\nabla_M + \bar\partial_{f^*M}$. Since φ is of type $(2,0)$, the components of types $(2,0)$, $(1,1)$, and $(0,2)$ of $d_{f^*\nabla_M}\eta = \varphi$ are

$$(f^*\nabla_M)\alpha = \varphi, \qquad \bar\partial_{f^*M}\alpha + (f^*\nabla_M)\beta = 0, \qquad \bar\partial_{f^*M}\beta = 0.$$

(1) Let $\mathbf{1} := (0, 1)$ be the smooth section of E which lifts the section 1 of $\mathcal{O}_S$. Since $\beta \wedge g$ is C^∞-linear in g, $\bar\partial_E$ satisfies the Leibniz rule. Moreover,

$$\bar\partial_E^2(u, g) = \big(\bar\partial_{f^*M}^2 u + (\bar\partial_{f^*M}\beta) \wedge g - \beta \wedge \bar\partial g + \beta \wedge \bar\partial g, \bar\partial^2 g\big) = \big((\bar\partial_{f^*M}\beta) \wedge g, 0\big) = 0.$$

Thus $\bar\partial_E$ defines a structure of holomorphic vector bundle on E [Huy05, Theorem 2.6.26], of rank $261 + 1 = 262$. Since $\bar\partial_E(u, 0) = (\bar\partial_{f^*M} u, 0)$ and the second component of $\bar\partial_E(u, g)$ is $\bar\partial g$, the inclusion $u \to (u, 0)$ and the projection $(u, g) \to g$ commute with the $(0,1)$-operators, hence are holomorphic. Finally, $\bar\partial_E \mathbf{1} = (\beta, 0)$. Since the projection $E \to \mathcal{O}_S$ is a surjective holomorphic map of vector bundles, there exist an open cover $\{U_i\}$ of S and holomorphic sections s_i of $E|_{U_i}$ which lift 1. The class of the extension, that is, the image of 1 under the connecting map $H^0(S, \mathcal{O}_S) \to H^1(S, f^*M)$ [Har77, III, Exercise 6.1 and Proposition 6.3(c)], is given by the Čech cocycle $(s_j - s_i)$ (Čech description of the connecting map, [GH78, Chapter 0, §3, p. 40] and [Har77, III, Lemma 4.4 and Exercise 4.4]). Let $\mathcal{Z}$ be the sheaf of $\bar\partial_{f^*M}$-closed f^*M-valued $(0,1)$-forms. We identify $H^{0,1}(S, f^*M)$ with $H^1(S, f^*M)$ by sending the Dolbeault class of a global section γ of $\mathcal{Z}$ to the Čech class of $(v_i - v_j)$, where $\{V_i\}$ is an open cover of S and v_i is a smooth section of f^*M on V_i with $\bar\partial_{f^*M} v_i = \gamma$. The identification is the negative of the Dolbeault isomorphism in degree one, which is induced by the connecting map of $0 \to f^*M \to A^0(f^*M) \to \mathcal{Z} \to 0$ [GH78, Chapter 0, §3, p. 40 and Chapter 1, §2, pp. 150–151]. The sections $v_i := \mathbf{1} - s_i$ of f^*M on U_i satisfy $\bar\partial_{f^*M} v_i = \beta$ and $v_i - v_j = s_j - s_i$. Hence the class of the extension is represented by β.

(2) Since ∇_M is flat (Construction-Proposition 4.1), so is $f^*\nabla_M$, and by [Del70, Chapter I, Théorème 2.17], f^*M has local bases consisting of $f^*\nabla_M$-horizontal holomorphic sections. Since $d_{f^*\nabla_M}$ is the componentwise exterior derivative in such a basis, $d_{f^*\nabla_M}^2 = 0$. Hence

$$d_\eta^2(u, g) = \big(d_{f^*\nabla_M}^2 u + (d_{f^*\nabla_M}\eta) \wedge g - \eta \wedge dg + \eta \wedge dg, d^2 g\big) = (\varphi \wedge g, 0),$$

that is,

$$d_\eta^2 = \begin{pmatrix} 0 & \varphi \\ 0 & 0 \end{pmatrix},$$

since the diagonal blocks are flat and the square of the strictly upper triangular part is zero. The operator d_η satisfies the Leibniz rule with respect to d, and the $(0,1)$ part of d_η is $\bar\partial_E$. Let ∇ be the $(1,0)$ part of d_η. For a local holomorphic section s of E, we have $\bar\partial_E s = 0$, so the $(1,1)$ part of $d_\eta^2 s$ is

$$\bar\partial_E(\nabla s) + \nabla(\bar\partial_E s) = \bar\partial_E(\nabla s),$$

which vanishes because d_η^2 is of type $(2,0)$. Thus ∇s is a holomorphic E-valued 1-form, and ∇ is a holomorphic connection on E with $d_\nabla = \nabla + \bar\partial_E = d_\eta$. By Definition 2.1, any holomorphic connection ∇' on E with $d_{\nabla'} = d_\eta$ is the $(1,0)$ part of d_η, so ∇ is unique. Since d_η maps $(u,0)$ to $(d_{f^*\nabla_M} u, 0)$, and the second component of $d_\eta(u,g)$ is dg, ∇ preserves f^*M and induces $f^*\nabla_M$ on f^*M and d on $\mathcal{O}_S$. For a local holomorphic section s of E,

$$R_\nabla(s) = \nabla(\nabla s) = \nabla(\nabla s) + \bar\partial_E(\nabla s) = d_\eta(\nabla s) = d_\eta(d_\eta s) = d_\eta^2 s.$$

Hence

$$R_\nabla = \begin{pmatrix} 0 & \varphi \\ 0 & 0 \end{pmatrix},$$

which is non-zero because $\varphi \neq 0$. □

5. Non-triviality on the universal cover

The goal of this section is to prove Proposition 5.2, which turns the condition $R_\nabla \neq 0$ into a holomorphic obstruction: $\mathbb{P}(q^*E)$ is non-trivial over $\widetilde{S}$. The obstructions in Theorem 1.4(2) and in Corollary 1.5 both come from Proposition 5.2. Corollary 1.5 needs non-triviality on the universal cover, and the absence of flat holomorphic connections alone is not enough for it (for instance, for a point x of a smooth fiber F_c of f, the line bundle $\mathcal{O}_{F_c}(x)$ admits no holomorphic connection since it has degree one [Ati57, Theorem 4, pp. 192–193], but the pullback of $\mathcal{O}_{F_c}(x)$ to the universal cover of F_c, which is $\mathbb{C}$ [GH78, Chapter 2, §2, p. 238], is trivial, as is any holomorphic line bundle on $\mathbb{C}$ [GH78, Chapter 0, §3, pp. 46–47]).

Lemma 5.1. *Let F be a compact connected Riemann surface, let $n \geq 1$ be an integer, and let G be a holomorphic vector bundle on F which is a non-split extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus n}$. Then $\mathbb{P}(G)$ is not isomorphic to $F \times \mathbb{P}^n$ over F.*

Proof. Let $u : G \to \mathcal{O}_F$ be the quotient map of the extension. Suppose that there exists an isomorphism $\Phi : \mathbb{P}(G) \to F \times \mathbb{P}^n$ over F. Holomorphic automorphisms of $\mathbb{P}^n$ are algebraic [Ser56, n° 19, Proposition 15], so $\mathrm{Aut}(\mathbb{P}^n) = \mathrm{PGL}_{n+1}(\mathbb{C})$ [Har77, II, Example 7.1.1]. Let $W \subset F$ be an open subset over which G is trivial. Then the restriction of Φ over W is given by a holomorphic map $W \to \mathrm{PGL}_{n+1}(\mathbb{C})$, since an element of $\mathrm{PGL}_{n+1}(\mathbb{C})$ is determined by the images of $n+2$ points in general position. On small open subsets of W, the map $W \to \mathrm{PGL}_{n+1}(\mathbb{C})$ lifts to a holomorphic map to $\mathrm{GL}_{n+1}(\mathbb{C})$, since the quotient map from $\mathrm{GL}_{n+1}(\mathbb{C})$ to $\mathrm{PGL}_{n+1}(\mathbb{C})$ has local holomorphic sections.

Thus there exist an open cover $\{W_i\}$ of F and isomorphisms $\psi_i : G|_{W_i} \to W_i \times \mathbb{C}^{n+1}$ such that the isomorphism $\mathbb{P}(G|_{W_i}) \to W_i \times \mathbb{P}^n$ induced by ψ_i is the restriction of Φ. For any i and j, the automorphism $\psi_i \circ \psi_j^{-1}$ of $(W_i \cap W_j) \times \mathbb{C}^{n+1}$ induces the identity on $(W_i \cap W_j) \times \mathbb{P}^n$, so

$$\psi_i \circ \psi_j^{-1} = \lambda_{ij} \cdot \mathrm{id}_{(W_i \cap W_j) \times \mathbb{C}^{n+1}}$$

for a nowhere vanishing holomorphic function λ_{ij} on $W_i \cap W_j$. The functions λ_{ij} satisfy the cocycle condition. Let $\mathcal{N}$ be the line bundle on F with transition functions λ_{ij}. Then $G \cong \mathcal{N}^{\oplus(n+1)}$.

Since G is an extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus n}$, $\det G \cong \mathcal{O}_F$. Thus

$$(n+1)\deg \mathcal{N} = \deg \det G = 0,$$

so $\deg \mathcal{N} = 0$. Since $n \geq 1$, the subbundle $\mathcal{O}_F^{\oplus n} \subset G$ implies $H^0(F, G) \neq 0$, hence $H^0(F, \mathcal{N}) \neq 0$. The divisor D of a non-zero section of $\mathcal{N}$ satisfies $D \geq 0$ and $\deg D = \deg \mathcal{N} = 0$, hence $D = 0$, and therefore $\mathcal{N} \cong \mathcal{O}_F$. Thus $G \cong \mathcal{O}_F^{\oplus(n+1)}$.

In a trivialization $G \cong \mathcal{O}_F^{\oplus(n+1)}$, the map u is a row $(u_0, \ldots, u_n)$ of holomorphic functions on F. Since F is compact and connected, $u_0, \ldots, u_n$ are constant, and $(u_0, \ldots, u_n) \neq 0$ since u is surjective. Let $v \in \mathbb{C}^{n+1}$ be a constant column vector with $\sum_{k=0}^n u_k v_k = 1$. Then v defines a map $\mathcal{O}_F \to G$ with $u \circ v = \mathrm{id}_{\mathcal{O}_F}$, so the extension splits, a contradiction. □

Proposition 5.2. *Notation and conditions as in Construction-Proposition 4.5. Then there exists $c \in C$ such that $F := F_c$ is a smooth elliptic curve, $E|_F$ is a non-split extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus 261}$, and $\mathbb{P}(E|_F)$ is not isomorphic to $F \times \mathbb{P}^{261}$ over F. Moreover, $\mathbb{P}(q^*E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the extension class of E does not come from C.

Since C, S, and the fibers F_c are projective, M is algebraic [Ser56, n° 20, Proposition 18], and the analytic cohomology of a coherent sheaf on them coincides with the algebraic cohomology [Ser56, n° 12, Théorème 1], and the isomorphism between them commutes with homomorphisms of sheaves, hence with restriction to F_c [Ser56, n° 11]. Thus we regard the class $e \in H^1(S, f^*M)$ of the extension $0 \to f^*M \to E \to \mathcal{O}_S \to 0$ as an algebraic cohomology class, and in **Steps 1 and 2** we use the Leray spectral sequence, the projection formula, and [Har77, III, Theorem 12.11] in the algebraic category.

By the projection formula [Har77, III, Exercise 8.3] and Lemma 3.5,

$$f_*f^*M \cong M, \qquad R^1f_*f^*M \cong M \otimes \mathcal{A}^{-1}.$$

Since C is a curve, $H^2(C, M) = 0$. Thus the five-term exact sequence of the Leray spectral sequence [Wei94, 5.8.6 and Exercise 5.1.3] for f and f^*M reads

$$0 \to H^1(C, M) \xrightarrow{f^*} H^1(S, f^*M) \to H^0(C, M \otimes \mathcal{A}^{-1}) \to 0.$$

Let $\sigma \in H^0(C, M \otimes \mathcal{A}^{-1})$ be the image of e.

Suppose that $\sigma = 0$. Then $e = f^*e'$ for some $e' \in H^1(C, M)$. Isomorphism classes of extensions of $\mathcal{O}$ by a locally free sheaf V correspond bijectively to elements of $\mathrm{Ext}^1(\mathcal{O}, V) = H^1(V)$ [Har77, III, Exercise 6.1 and Proposition 6.3(c)], the first map of the Leray sequence is the pullback $H^1(C, M) \to H^1(S, f^*M)$ (it is the edge map $H^1(C, f_*f^*M) \to H^1(S, f^*M)$ composed with $M \cong f_*f^*M$), and the class of the pullback of an extension is the pullback of the class (connecting homomorphisms are natural). Thus there exists an extension $0 \to M \to N \to \mathcal{O}_C \to 0$ on C such that E is isomorphic to f^*N as an extension. Let ∇' be the holomorphic connection on f^*N corresponding to ∇ under an isomorphism $E \cong f^*N$. By Corollary 3.7, ∇' is flat, so $R_{\nabla} = 0$. This contradicts $R_{\nabla} \neq 0$ (Construction-Proposition 4.5(2)). Hence $\sigma \neq 0$.

Step 2. In this step we find a smooth fiber F_c such that $E|_{F_c}$ is a non-split extension.

Since S is integral and f is surjective, f is flat [Har77, III, Proposition 9.7], so f^*M is flat over C. For any $c \in C$, since F_c is a curve, $H^2(F_c, f^*M|_{F_c}) = 0$, so [Har77, III, Theorem 12.11(a)] implies $(R^2f_*f^*M) \otimes k(c) = 0$, hence $R^2f_*f^*M = 0$ (by Nakayama's lemma), which is locally free. By [Har77, III, Theorem 12.11(b)] the base change map

$$\kappa_c : (R^1f_*f^*M) \otimes k(c) \to H^1(F_c, f^*M|_{F_c}) = H^1(F_c, \mathcal{O}_{F_c}) \otimes M(c),$$

where $M(c) := M \otimes k(c)$ is the fiber of M at c, is surjective, hence an isomorphism by [Har77, III, Theorem 12.11(a)].

The edge map $H^1(S, f^*M) \to H^0(C, R^1f_*f^*M)$ sends a class to the section given by the restrictions of the class to $H^1(f^{-1}(W), f^*M)$ for open subsets $W \subset C$, and κ_c is induced by the restriction maps $H^1(f^{-1}(W), f^*M) \to H^1(F_c, f^*M|_{F_c})$ for $c \in W$. Thus the composite of the edge map, the value at c, and κ_c is the restriction $H^1(S, f^*M) \to H^1(F_c, f^*M|_{F_c})$. Since extension classes commute with restriction, $\kappa_c(\sigma(c))$ is the class of the extension

$$0 \to f^*M|_{F_c} \to E|_{F_c} \to \mathcal{O}_{F_c} \to 0.$$

Since σ is a non-zero section of the vector bundle $M \otimes \mathcal{A}^{-1}$ on the connected curve C, the set $\{c \in C \mid \sigma(c) = 0\}$ is finite. By Lemma 3.3(2), exactly twenty fibers of f are singular. Thus there exists $c \in C$ such that $\sigma(c) \neq 0$ and F_c is smooth. Let $F := F_c$. By Lemma 3.3(2), F is a smooth plane cubic curve, hence a smooth elliptic curve. Moreover $f^*M|_F \cong \mathcal{O}_F^{\oplus 261}$, and since κ_c is injective, $E|_F$ is a non-split extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus 261}$.

Step 3. In this step we show that $\mathbb{P}(E|_F)$ is non-trivial. By Lemma 5.1 with $n = 261$ and $G = E|_F$, $\mathbb{P}(E|_F)$ is not isomorphic to $F \times \mathbb{P}^{261}$ over F.

Step 4. In this step we show that $\mathbb{P}(q^*E)$ is non-trivial. Let $\tilde{c} \in \widetilde{C}$ be a point over c, and let $\widetilde{F} \subset \widetilde{S}$ be the fiber of the projection $\widetilde{S} \to \widetilde{C}$ over $\tilde{c}$. By Lemma 3.4, $q|_{\widetilde{F}} : \widetilde{F} \to F$ is an isomorphism, and

$$\mathbb{P}(q^*E)|_{\widetilde{F}} \cong \mathbb{P}\big((q^*E)|_{\widetilde{F}}\big) \cong \mathbb{P}(E|_F)$$

over $q|_{\widetilde{F}}$. Suppose that there exists an isomorphism $\mathbb{P}(q^*E) \to \widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$. Restricting it to $\widetilde{F}$ and composing with $q|_{\widetilde{F}}$, we obtain an isomorphism $\mathbb{P}(E|_F) \to F \times \mathbb{P}^{261}$ over F, which contradicts **Step 3**. Hence $\mathbb{P}(q^*E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$. □

6. Split tangent bundles of fiber bundles

The goal of this section is to prove two general facts which, together with a dimension count in Section 7, turn Theorem 1.4 into a counterexample to Conjecture 1.3. First, a holomorphic connection on a vector bundle splits the tangent bundle of the associated projective bundle, and the horizontal summand is not integrable when a value of the curvature is not scalar (Definition-Lemma 6.1 and Lemma 6.2). Second, for a fiber bundle whose fiber is compact, connected, simply connected, and without holomorphic 1-forms, a product decomposition of the universal cover as in [Hör08, Conjecture 1.6], or one whose second factor has the dimension of the base and whose second projection is constant on the fibers, is compatible with the fibration (Proposition 6.3).

6.1. Horizontal subbundles.

In this subsection, we construct the horizontal subbundle of a holomorphic connection and compute the Lie bracket of two horizontal vector fields.

Definition-Lemma 6.1. Let Z be a complex manifold, let E be a holomorphic vector bundle of rank $n+1 \geq 2$ on Z with a holomorphic connection ∇, let $X := \mathbb{P}(E)$, and let $p : X \to Z$ be the projection. Then there exists a unique holomorphic subbundle $H_\nabla \subset T_X$ with the following property: for any open subset $W \subset Z$ and any isomorphism $E|_W \cong \mathcal{O}_W^{\oplus(n+1)}$ in which $\nabla = d + \Gamma$ for a matrix Γ of holomorphic 1-forms on W, and for any $z \in W$ and $v \in \mathbb{C}^{n+1}\backslash\{0\}$, the fiber of H_∇ at the point $[v] \in \mathbb{P}(E_z)$ is the image of $\{(\xi, -\Gamma(\xi)v) \mid \xi \in T_{Z,z}\}$ under the differential at (z, v) of the quotient map $W \times (\mathbb{C}^{n+1}\backslash\{0\}) \to p^{-1}(W)$. Moreover, $T_X = T_{X/Z} \oplus H_\nabla$, where $T_{X/Z} := \ker(dp)$ is the relative tangent bundle of p, and dp restricts to an isomorphism $H_\nabla \to p^*T_Z$. We call H_∇ the *horizontal subbundle* of ∇.

Proof. Let $W \subset Z$ be an open subset with an isomorphism $E|_W \cong \mathcal{O}_W^{\oplus(n+1)}$ in which $\nabla = d + \Gamma$, and let $\rho : W \times (\mathbb{C}^{n+1}\backslash\{0\}) \to p^{-1}(W)$ be the quotient map. We regard the tangent space of $W \times (\mathbb{C}^{n+1}\backslash\{0\})$ at (z, v) as $T_{Z,z} \oplus \mathbb{C}^{n+1}$, and for $\xi \in T_{Z,z}$ we write $\Gamma(\xi) \in \mathrm{End}(\mathbb{C}^{n+1})$ for the value of Γ at ξ.

For $\lambda \in \mathbb{C}^*$, let $m_\lambda(z, v) := (z, \lambda v)$. Then $\rho \circ m_\lambda = \rho$ and $dm_\lambda(\xi, w) = (\xi, \lambda w)$, so

$$d\rho_{(z,\lambda v)}(\xi, -\Gamma(\xi)\lambda v) = d\rho_{(z,\lambda v)}\big(dm_\lambda(\xi, -\Gamma(\xi)v)\big) = d\rho_{(z,v)}(\xi, -\Gamma(\xi)v).$$

Thus, for a point $x = [v] \in p^{-1}(W)$ over z, the linear map

$$\ell_x : T_{Z,z} \to T_{X,x}, \quad \xi \to d\rho_{(z,v)}(\xi, -\Gamma(\xi)v)$$

does not depend on the representative v of x. The maps ℓ_x form a morphism of vector bundles $\ell_W : (p^*T_Z)|_{p^{-1}(W)} \to T_X|_{p^{-1}(W)}$. Since $(z, v, \xi) \to (\xi, -\Gamma(\xi)v)$ is holomorphic and ρ admits local holomorphic sections, ℓ_W is holomorphic. Since $p \circ \rho$ is the projection to W, we have $dp \circ \ell_W = \mathrm{id}_{(p^*T_Z)|_{p^{-1}(W)}}$.

We show that ℓ_W does not depend on the trivialization. Let $W' \subset Z$ be another open subset with an isomorphism $E|_{W'} \cong \mathcal{O}_{W'}^{\oplus(n+1)}$ in which $\nabla = d + \Gamma'$, and let ρ' and $\ell_{W'}$ be the corresponding maps. Let $a : W \cap W' \to \mathrm{GL}_{n+1}(\mathbb{C})$ be the holomorphic map such that a local

section of E with coordinate vector s' in the second trivialization has coordinate vector as' in the first. Since $\nabla s'$ has coordinate vector $a^{-1}(d(as')+\Gamma as')$ in the second trivialization, we have

$$\Gamma' = a^{-1}\Gamma a + a^{-1}da.$$

Moreover, $\rho'(z,v) = \rho(z,a(z)v)$ on $(W\cap W')\times(\mathbb{C}^{n+1}\backslash\{0\})$. The differential at (z,v) of the map

$$(z,v)\to(z,a(z)v)$$

sends (ξ,w) to $(\xi, da(\xi)v + a(z)w)$. For $w=-\Gamma'(\xi)v$ we obtain

$$da(\xi)v - a(z)\big(a(z)^{-1}\Gamma(\xi)a(z) + a(z)^{-1}da(\xi)\big)v = -\Gamma(\xi)a(z)v.$$

Thus

$$d\rho'_{(z,v)}(\xi,-\Gamma'(\xi)v) = d\rho_{(z,a(z)v)}\big(\xi,-\Gamma(\xi)a(z)v\big),$$

that is, $\ell_W = \ell_{W'}$ on $p^{-1}(W\cap W')$.

Hence the morphisms ℓ_W glue to a holomorphic morphism $\ell: p^*T_Z\to T_X$ with $dp\circ\ell = \mathrm{id}_{p^*T_Z}$. Let $H_\nabla := \ell(p^*T_Z)$. Since ℓ has the left inverse dp, H_∇ is a holomorphic subbundle of T_X, and H_∇ has the property in the statement by construction. Since any point of X is contained in $p^{-1}(W)$ for some open subset $W\subset Z$ with an isomorphism $E|_W\cong\mathcal{O}_W^{\oplus(n+1)}$, the property in the statement determines the fibers of H_∇ at all points of X, so H_∇ is unique. Finally, $dp\circ\ell=\mathrm{id}_{p^*T_Z}$ implies that $T_X = \ker(dp)\oplus\ell(p^*T_Z) = T_{X/Z}\oplus H_\nabla$, and that $dp|_{H_\nabla}$ is the inverse of the isomorphism $\ell: p^*T_Z\to H_\nabla$. □

Lemma 6.2. *Notation as in Definition-Lemma 6.1. Assume that there exist $z\in Z$ and $\xi,\zeta\in T_{Z,z}$ such that $R_\nabla(\xi,\zeta)\in\mathrm{End}(E_z)$ is not a scalar multiple of id_{E_z}. Then H_∇ is not integrable.*

Proof. Recall that H_∇ is integrable if the Lie bracket of any two local holomorphic sections of H_∇ is a section of H_∇. We construct two local sections of H_∇ whose Lie bracket is not a section of H_∇.

Let W be a coordinate neighborhood of z with coordinates $t_1,\dots,t_k$ and with an isomorphism $E|_W\cong\mathcal{O}_W^{\oplus(n+1)}$ in which $\nabla = d+\Gamma$. Let $\rho: W\times(\mathbb{C}^{n+1}\backslash\{0\})\to p^{-1}(W)$ be the quotient map. Let ξ' and ζ' be the vector fields on W with constant coefficients in $t_1,\dots,t_k$ such that $\xi'(z)=\xi$ and $\zeta'(z)=\zeta$. Then $[\xi',\zeta']=0$. Let $\Gamma_1:=\Gamma(\xi')$ and $\Gamma_2:=\Gamma(\zeta')$, which are matrices of holomorphic functions on W.

We evaluate 2-forms by the convention $(\alpha\wedge\beta)(\xi,\zeta)=\alpha(\xi)\beta(\zeta)-\alpha(\zeta)\beta(\xi)$ for 1-forms α and β. A direct calculation from Definition 2.1 shows that $R_\nabla = d\Gamma+\Gamma\wedge\Gamma$ on W. Since $[\xi',\zeta']=0$, we obtain

$$R_\nabla(\xi',\zeta') = \xi'(\Gamma_2) - \zeta'(\Gamma_1) + \Gamma_1\Gamma_2 - \Gamma_2\Gamma_1.$$

Let $\widehat{L}_1$ and $\widehat{L}_2$ be the holomorphic vector fields on $W\times(\mathbb{C}^{n+1}\backslash\{0\})$ defined by

$$\widehat{L}_1(t,v) := \big(\xi'(t),-\Gamma_1(t)v\big),\qquad \widehat{L}_2(t,v) := \big(\zeta'(t),-\Gamma_2(t)v\big).$$

Since the first components of $\widehat{L}_1$ and $\widehat{L}_2$ depend only on t and commute, the first component of $[\widehat{L}_1,\widehat{L}_2]$ is $[\xi',\zeta']=0$, and the second component is $\widehat{L}_1(-\Gamma_2 v)-\widehat{L}_2(-\Gamma_1 v)$. We have

$$\widehat{L}_1(-\Gamma_2 v) = -\xi'(\Gamma_2)v + \Gamma_2\Gamma_1 v,\qquad \widehat{L}_2(-\Gamma_1 v) = -\zeta'(\Gamma_1)v + \Gamma_1\Gamma_2 v.$$

Thus

$$[\widehat{L}_1,\widehat{L}_2](t,v) = \big(0,-R_\nabla(\xi',\zeta')(t)v\big).$$

A different sign convention for the bracket or for R_∇ changes the right-hand side by a non-zero constant factor, which does not affect the argument below (cf. [KN63, Chapter II, Corollary 5.3] for principal bundles).

Since dp restricts to an isomorphism $H_\nabla\to p^*T_Z$ (Definition-Lemma 6.1), there exist unique holomorphic sections L_1 and L_2 of H_∇ over $p^{-1}(W)$ with $dp(L_1)=p^*\xi'$ and $dp(L_2)=p^*\zeta'$. By Definition-Lemma 6.1, $d\rho(\widehat{L}_1(t,v))$ is a vector of H_∇ at $\rho(t,v)$, and $dp(d\rho(\widehat{L}_1(t,v))) = \xi'(t)$.

Hence $d\rho(\widehat{L}_1(t,v)) = L_1(\rho(t,v))$, and the argument with ζ' and L_2 in place of ξ' and L_1 shows $d\rho(\widehat{L}_2(t,v)) = L_2(\rho(t,v))$. Since the Lie bracket of ρ-related vector fields is ρ-related,

$$[L_1, L_2](\rho(t,v)) = d\rho_{(t,v)}\big(0, -R_\nabla(\xi',\zeta')(t)v\big).$$

The restriction of ρ to $\{z\} \times (\mathbb{C}^{n+1}\backslash\{0\})$ is the quotient map onto $p^{-1}(z)$, whose differential at v is surjective onto $T_{p^{-1}(z),[v]}$ with kernel $\mathbb{C}v$. Since $R_\nabla(\xi',\zeta')(z) = R_\nabla(\xi,\zeta)$, the vector $[L_1,L_2]([v])$ is contained in $T_{X/Z}$, and it is zero if and only if $R_\nabla(\xi,\zeta)v \in \mathbb{C}v$. As $T_{X/Z} \cap H_\nabla = 0$, we have $[L_1,L_2]([v]) \in H_\nabla$ if and only if $R_\nabla(\xi,\zeta)v \in \mathbb{C}v$.

Suppose that H_∇ is integrable. Then $R_\nabla(\xi,\zeta)v \in \mathbb{C}v$ for any $v \in \mathbb{C}^{n+1}\backslash\{0\}$. Let $T := R_\nabla(\xi,\zeta)$, regarded as a matrix via the trivialization. If $v, w \in \mathbb{C}^{n+1}$ are linearly independent and $Tv = av$, $Tw = bw$, and $T(v+w) = c(v+w)$, then $(a-c)v + (b-c)w = 0$, so $a = b = c$. Since $n+1 \geq 2$, for any two non-zero vectors $u, u' \in \mathbb{C}^{n+1}$ there exists $a \in \mathbb{C}$ with $Tu = au$ and $Tu' = au'$, either because u and u' are proportional or by the previous sentence. Thus T is a scalar multiple of id_{E_z}, a contradiction. □

6.2. **Products on the universal cover.** In this subsection, we prove Proposition 6.3. It reduces the test of [Hör08, Conjecture 1.6] on $\mathbb{P}(E)$ to the question whether a fiber bundle over the universal cover of the base is trivial, and it reduces the test of Conjecture 1.3 on $\mathbb{P}(E)$ to that question for product decompositions whose second projection is constant on the fibers. For the $\mathbb{P}^{261}$-bundle of Corollary 1.5, Section 7 shows by a dimension count that the second projection of any product decomposition of the universal cover into factors of dimensions 261 and 2 is constant on the fibers.

Proposition 6.3. *Let X and Z be connected complex manifolds, let F be a compact connected simply connected complex manifold with $\dim F > 0$ and $H^0(F, \Omega^1_F) = 0$, and let $p : X \to Z$ be a holomorphic fiber bundle with fiber F (see Definition 2.2). Assume that $\dim Z > 0$ and that $T_X = T_{X/Z} \oplus H$ for a holomorphic subbundle $H \subset T_X$. Let $g : \widetilde{Z} \to Z$ be the universal cover, let $\widetilde{X} := X \times_Z \widetilde{Z}$, and let $\pi : \widetilde{X} \to X$ and $\widetilde{p} : \widetilde{X} \to \widetilde{Z}$ be the projections. Then π is a universal cover of X, and the following are equivalent.*

(1) There exist connected simply connected complex manifolds Y_1, Y_2 and a biholomorphism $\Phi : \widetilde{X} \to Y_1 \times Y_2$ such that $\pi^* T_{X/Z} \cong \Phi^* \operatorname{pr}_1^* T_{Y_1}$ and $\pi^* H \cong \Phi^* \operatorname{pr}_2^* T_{Y_2}$ as holomorphic vector bundles.

(2) There exists a biholomorphism $\widetilde{X} \to F \times \widetilde{Z}$ over $\widetilde{Z}$.

Moreover, let Y_1, Y_2 be connected simply connected complex manifolds and $\Phi : \widetilde{X} \to Y_1 \times Y_2$ a biholomorphism, and assume that one of the following holds:

(a) $\pi^* H \cong \Phi^* \operatorname{pr}_2^* T_{Y_2}$ as holomorphic vector bundles;

(b) $\dim Y_2 = \dim Z$, and $\operatorname{pr}_2 \circ \Phi$ is constant on any fiber of $\widetilde{p}$.

Then there exists a biholomorphism $h : \widetilde{Z} \to Y_2$ such that $\operatorname{pr}_2 \circ \Phi = h \circ \widetilde{p}$, and (2) holds.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that π is a universal cover of X, and that $d\widetilde{p}$ restricts to an isomorphism from $\pi^* H$ onto $\widetilde{p}^* T_{\widetilde{Z}}$.

Let $W \subset Z$ be a connected open subset which is evenly covered by g and over which there exists a biholomorphism $p^{-1}(W) \cong W \times F$ over W. Any connected component $\widetilde{W}$ of $g^{-1}(W)$ is mapped biholomorphically onto W by g, and $\widetilde{p}^{-1}(\widetilde{W}) = p^{-1}(W) \times_W \widetilde{W}$. Thus π maps $\widetilde{p}^{-1}(\widetilde{W})$ biholomorphically onto $p^{-1}(W)$, and $\widetilde{p}^{-1}(\widetilde{W}) \cong \widetilde{W} \times F$ over $\widetilde{W}$. Since such open subsets W cover Z, π is a holomorphic covering map, and $\widetilde{p}$ is a holomorphic fiber bundle with fiber F.

We show that $\widetilde{X}$ is connected. Let X' be a connected component of $\widetilde{X}$. Since F is connected, any fiber of $\widetilde{p}$ is contained in a single connected component of $\widetilde{X}$. Since any point of $\widetilde{Z}$ has a connected open neighborhood $\widetilde{W}$ with $\widetilde{p}^{-1}(\widetilde{W}) \cong \widetilde{W} \times F$ over $\widetilde{W}$, and $\widetilde{W} \times F$ is connected, the

set of points $z \in \widetilde{Z}$ with $\widetilde{p}^{-1}(z) \subset X'$ and the set of points $z \in \widetilde{Z}$ with $\widetilde{p}^{-1}(z) \cap X' = \emptyset$ are both open. The first set is non-empty and $\widetilde{Z}$ is connected, so $X' = \widetilde{X}$.

By [Hat02, Proposition 4.48 and Theorem 4.41], applied to the fiber bundle $\widetilde{p}$, the sequence

$$\pi_1(F) \to \pi_1(\widetilde{X}) \to \pi_1(\widetilde{Z})$$

is exact. Since F and $\widetilde{Z}$ are simply connected, $\pi_1(\widetilde{X}) = 1$. Thus π is a universal cover of X.

Since p is locally the projection $W \times F \to W$, $dp : T_X \to p^*T_Z$ is surjective with kernel $T_{X/Z}$. As $T_X = T_{X/Z} \oplus H$, dp restricts to an isomorphism $H \to p^*T_Z$. Since π is a local biholomorphism, we regard π^*T_X as $T_{\widetilde{X}}$ via $d\pi$, so that $\pi^*T_{X/Z}$ and π^*H are subbundles of $T_{\widetilde{X}}$. From $p \circ \pi = g \circ \widetilde{p}$ we obtain $\pi^*(dp) = \widetilde{p}^*(dg) \circ d\widetilde{p}$ as morphisms $T_{\widetilde{X}} \to \widetilde{p}^*g^*T_Z$, and $dg : T_{\widetilde{Z}} \to g^*T_Z$ is an isomorphism. Hence $\ker(d\widetilde{p}) = \pi^*T_{X/Z}$, and

$$d\widetilde{p}|_{\pi^*H} = \widetilde{p}^*(dg)^{-1} \circ \pi^*(dp|_H) : \pi^*H \to \widetilde{p}^*T_{\widetilde{Z}}$$

is an isomorphism of holomorphic vector bundles.

Step 2. In this step we show that (a) implies (b).

Assume (a). Comparing ranks with **Step 1**, we have $\dim Y_2 = \dim Z$. Let $f := \mathrm{pr}_2 \circ \Phi : \widetilde{X} \to Y_2$.

Fix $z \in \widetilde{Z}$, let $\widetilde{X}_z := \widetilde{p}^{-1}(z)$, and let $f_z := f|_{\widetilde{X}_z}$. Since $\widetilde{X}_z \cong F$, $\widetilde{X}_z$ is compact and connected, and $H^0(\widetilde{X}_z, \Omega^1_{\widetilde{X}_z}) = 0$. By **Step 1**,

$$f_z^*T_{Y_2} = (\Phi^* \mathrm{pr}_2^* T_{Y_2})|_{\widetilde{X}_z} \cong (\pi^*H)|_{\widetilde{X}_z} \cong (\widetilde{p}^*T_{\widetilde{Z}})|_{\widetilde{X}_z} \cong \mathcal{O}_{\widetilde{X}_z}^{\oplus \dim Z}.$$

The differential df_z is a holomorphic section of

$$\Omega^1_{\widetilde{X}_z} \otimes f_z^*T_{Y_2} \cong (\Omega^1_{\widetilde{X}_z})^{\oplus \dim Z},$$

so $df_z = 0$. Thus f_z is locally constant, and f_z is constant since $\widetilde{X}_z$ is connected. Hence (b) holds.

Step 3. In this step we prove the last statement of the proposition, and that (1) implies (2).

By **Step 2**, we may assume (b). Let $f := \mathrm{pr}_2 \circ \Phi : \widetilde{X} \to Y_2$, and for $z \in \widetilde{Z}$ let $\widetilde{X}_z := \widetilde{p}^{-1}(z)$. Since $\dim Y_2 = \dim Z$, we have $\dim Y_1 = \dim \widetilde{X} - \dim Z = \dim F$.

Since f is constant on any fiber of $\widetilde{p}$, there exists a map $h : \widetilde{Z} \to Y_2$ with $f = h \circ \widetilde{p}$. By **Step 1**, $\widetilde{p}$ admits a holomorphic section σ over a neighborhood of any point of $\widetilde{Z}$, and $h = f \circ \sigma$ there, so h is holomorphic. Since Φ is a biholomorphism and pr_2 is a submersion, df is surjective at any point. From $df = dh \circ d\widetilde{p}$, dh is surjective at any point. As $\dim Y_2 = \dim \widetilde{Z}$, h is a local biholomorphism by the inverse function theorem.

For $z \in \widetilde{Z}$, let $\phi_z := \mathrm{pr}_1 \circ \Phi|_{\widetilde{X}_z} : \widetilde{X}_z \to Y_1$. Since $\Phi(\widetilde{X}_z) \subset Y_1 \times \{h(z)\}$ and pr_1 restricts to a biholomorphism $Y_1 \times \{h(z)\} \to Y_1$, the map ϕ_z is injective with injective differential. As $\dim \widetilde{X}_z = \dim Y_1$, ϕ_z is a local biholomorphism, so $\phi_z(\widetilde{X}_z)$ is open in Y_1. Since $\widetilde{X}_z$ is compact and Y_1 is Hausdorff, $\phi_z(\widetilde{X}_z)$ is closed in Y_1. Since $\phi_z(\widetilde{X}_z)$ is non-empty and Y_1 is connected, $\phi_z(\widetilde{X}_z) = Y_1$. Thus ϕ_z is a biholomorphism, and

$$\Phi(\widetilde{X}_z) = Y_1 \times \{h(z)\}. \tag{6.1}$$

Since Φ is surjective, $h(\widetilde{Z}) = f(\widetilde{X}) = Y_2$. If $h(z) = h(z')$ for points $z \neq z'$ of $\widetilde{Z}$, then $\Phi(\widetilde{X}_z) = \Phi(\widetilde{X}_{z'})$ by (6.1), while $\widetilde{X}_z \cap \widetilde{X}_{z'} = \emptyset$, which contradicts the injectivity of Φ. Thus h is bijective. A bijective local biholomorphism is a biholomorphism, so h is a biholomorphism with $\mathrm{pr}_2 \circ \Phi = h \circ \widetilde{p}$.

Fix $z_0 \in \widetilde{Z}$ and a biholomorphism $e : F \to \widetilde{X}_{z_0}$, which exists by **Step 1**. Then $\phi_{z_0} \circ e : F \to Y_1$ is a biholomorphism. Let

$$\Theta := \left((\phi_{z_0} \circ e)^{-1} \times h^{-1}\right) \circ \Phi : \widetilde{X} \to F \times \widetilde{Z}.$$

Then Θ is a biholomorphism, and $\mathrm{pr}_2 \circ \Theta = h^{-1} \circ f = \widetilde{p}$, where $\mathrm{pr}_2 : F \times \widetilde{Z} \to \widetilde{Z}$ is the projection. Thus (2) holds. Since (1) contains the isomorphism $\pi^* H \cong \Phi^* \mathrm{pr}_2^* T_{Y_2}$, which is (a), (1) implies (2).

Step 4. In this step we show that (2) implies (1).

Let $\Theta : \widetilde{X} \to F \times \widetilde{Z}$ be a biholomorphism with $\mathrm{pr}_2 \circ \Theta = \widetilde{p}$. The manifolds F and $\widetilde{Z}$ are connected and simply connected. Since Θ is a biholomorphism over $\widetilde{Z}$, $d\Theta$ maps $\ker(d\widetilde{p}) = \pi^* T_{X/Z}$ isomorphically onto $\Theta^* \ker(d\,\mathrm{pr}_2) = \Theta^* \mathrm{pr}_1^* T_F$. By **Step 1**,

$$\pi^* H \cong \widetilde{p}^* T_{\widetilde{Z}} = \Theta^* \mathrm{pr}_2^* T_{\widetilde{Z}}.$$

Thus $Y_1 := F$, $Y_2 := \widetilde{Z}$, and $\Phi := \Theta$ satisfy (1). □

7. Proofs of the main theorems

In this section, we prove Theorem 1.4 and Corollary 1.5.

Proof of Theorem 1.4. We use the notation of Construction 3.2 (smooth projective curves of genus two exist [Har77, IV, Exercise 2.2(b)], and one can take $\mathcal{A} = \mathcal{O}_C(5P)$ for a point $P \in C$), where the morphism $h : C \to \mathbb{P}^1$ exists by Lemma 3.1. By Lemma 3.3(1), S is a smooth connected projective surface. By Lemma 3.3(2), any fiber F_c is a plane cubic curve which is either smooth or the union of three lines in general position. Since any two curves in $\mathbb{P}^2$ intersect [Har77, I, Theorem 7.2], a smooth plane cubic curve is irreducible (a common point of two components is singular), and the union of three lines in general position is connected. Thus f is surjective, and any fiber of f is connected. By Lemma 3.4, $f_* : \pi_1(S) \to \pi_1(C)$ is an isomorphism, and $q : \widetilde{S} \to S$ is a universal cover of S. In particular, $\widetilde{S}$ is connected and simply connected.

By Lemma 2.5, there exist a line bundle $\mathcal{L}$ on C with $\mathcal{L}^{\otimes 2} \cong \omega_C$, a holomorphic vector bundle E_0 on C which is a non-split extension of $\mathcal{L}^{-1}$ by $\mathcal{L}$, and a holomorphic connection ∇_0 on E_0 which induces the trivial connection on $\det E_0 \cong \mathcal{O}_C$. Let $M := \mathrm{Sym}^{260} E_0$, and let ∇_M be the holomorphic connection on M induced by ∇_0. By Construction-Proposition 4.1, $\operatorname{rank} M = 261$, ∇_M is flat, and ∇_M induces the trivial connection on $\det M \cong \mathcal{O}_C$. By Proposition 4.4, there exist a non-zero section $\varphi \in H^0(S, \Omega_S^2 \otimes f^* M)$ and a smooth $f^* M$-valued 1-form η on S such that $d_{f^*\nabla_M} \eta = \varphi$. Let E and ∇ be the holomorphic vector bundle and the holomorphic connection defined by φ and η in Construction-Proposition 4.5. The arguments below use only that $\varphi \neq 0$ and $d_{f^*\nabla_M} \eta = \varphi$, so they show that (1)–(5) hold for the holomorphic vector bundle E and the holomorphic connection ∇ given by Construction-Proposition 4.5 for any such choices of $\mathcal{L}$, E_0, ∇_0, φ, and η. By Construction-Proposition 4.5(1), E is a holomorphic vector bundle of rank 262 which fits into an exact sequence of holomorphic vector bundles

$$0 \to f^* M \to E \to \mathcal{O}_S \to 0.$$

By Construction-Proposition 4.5(2), ∇ preserves $f^* M$, induces $f^* \nabla_M$ on $f^* M$ and d on $\mathcal{O}_S$, and

$$R_\nabla = \begin{pmatrix} 0 & \varphi \\ 0 & 0 \end{pmatrix} \neq 0.$$

This proves (1).

By Proposition 5.2, there exists $c \in C$ such that $F := F_c$ is a smooth elliptic curve, $E|_F$ is a non-split extension of $\mathcal{O}_F$ by $\mathcal{O}_F^{\oplus 261}$, and $\mathbb{P}(E|_F)$ is not isomorphic to $F \times \mathbb{P}^{261}$ over F. Moreover, by Proposition 5.2, $\mathbb{P}(q^* E)$ is not isomorphic to $\widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$. This proves (4) and (5).

We prove (2). Suppose that E admits a flat holomorphic connection ∇'. Then $q^* \nabla'$ is a holomorphic connection on $q^* E$ with $R_{q^*\nabla'} = q^* R_{\nabla'} = 0$. Since $\widetilde{S}$ is simply connected, Lemma 2.3(1) implies that $q^* E \cong \mathcal{O}_{\widetilde{S}}^{\oplus 262}$. Thus $\mathbb{P}(q^* E) \cong \widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$, which contradicts (5). Hence E admits no flat holomorphic connection.

Let $p : \mathbb{P}(E) \to S$ be the projection, and suppose that $\mathbb{P}(E)$ admits a flat holomorphic connection. Then there exist an open cover $\{W_i\}$ of S and biholomorphisms

$$\psi_i : p^{-1}(W_i) \to W_i \times \mathbb{P}^{261}$$

over W_i such that any transition map $\psi_{ij} : W_i \cap W_j \to \mathrm{Aut}(\mathbb{P}^{261})$ is locally constant. Since $\mathbb{P}(q^*E) = \mathbb{P}(E) \times_S \widetilde{S}$, the base changes of the ψ_i along q are biholomorphisms from the inverse image of $q^{-1}(W_i)$ in $\mathbb{P}(q^*E)$ to $q^{-1}(W_i) \times \mathbb{P}^{261}$ over $q^{-1}(W_i)$, with transition maps $\psi_{ij} \circ q$. The maps $\psi_{ij} \circ q$ are locally constant, so $\mathbb{P}(q^*E)$ admits a flat holomorphic connection. By Lemma 2.3(2), $\mathbb{P}(q^*E) \cong \widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$, which contradicts (5). This proves (2).

We prove (3). By Construction-Proposition 4.5, the smooth complex vector bundle underlying E is $f^*M \oplus \mathcal{O}_S$. By the Whitney product formula [MS74, (14.7)] (and naturality),

$$c(E) = c(f^*M) \cdot c(\mathcal{O}_S) = f^*c(M).$$

Since $\det M \cong \mathcal{O}_C$, we have $c_1(M) = c_1(\det M) = 0$ (splitting principle), and $c_i(M) = 0$ for any $i \geq 2$ since $H^{2i}(C, \mathbb{Z}) = 0$. Thus $c(M) = 1$, and so $c_i(E) = 0$ in $H^{2i}(S, \mathbb{Z})$ for any $i \geq 1$. Moreover, $q^*f^*\nabla_M$ is a holomorphic connection on q^*f^*M, and it is flat since $R_{q^*f^*\nabla_M} = q^*f^*R_{\nabla_M} = 0$. Since $\widetilde{S}$ is simply connected, Lemma 2.3(1) implies that $q^*f^*M \cong \mathcal{O}_{\widetilde{S}}^{\oplus 261}$. Therefore, as smooth complex vector bundles,

$$q^*E \cong q^*f^*M \oplus \mathcal{O}_{\widetilde{S}} \cong \widetilde{S} \times \mathbb{C}^{262}.$$

This proves (3).

The surface S and the morphism h ($\deg h = \deg h^*\mathcal{O}_{\mathbb{P}^1}(1) = \deg \mathcal{A} = 5$ [Har77, II, Proposition 6.9]) are as in Construction 3.2, and $M = \mathrm{Sym}^{260} E_0$ for the extension E_0 of Lemma 2.5, so S, h, and M are as in the last part of Theorem 1.4. Since $h^*\mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{A}$, Lemma 3.3(1) implies that S is a smooth member of $|\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes h^*\mathcal{O}_{\mathbb{P}^1}(1)|$ on $\mathbb{P}^2 \times C$. □

Proof of Corollary 1.5. We use the notation of Construction 3.2 and Construction-Proposition 4.1. Let $\mathcal{L}$, E_0, ∇_0, M, and ∇_M be as in Construction-Proposition 4.1, which exist by Lemma 2.5, let φ and η be as in Construction-Proposition 4.5, which exist by Proposition 4.4, and let E and ∇ be the holomorphic vector bundle and the holomorphic connection given by Construction-Proposition 4.5. The argument below holds for any such choices. Let $X := \mathbb{P}(E)$, and let $p : X \to S$ be the projection.

By GAGA [Ser56, n° 20, Proposition 18], E is the holomorphic vector bundle associated with an algebraic vector bundle on the projective surface S. Since $\mathbb{P}(E)$ is the bundle of lines in E, which is $\mathbb{P}(E^\vee)$ in the notation of [Har77, II, §7, the definition of $\mathbb{P}(\mathcal{E})$ and Proposition 7.12], X is projective by [Har77, II, Proposition 7.10(b) and Exercise 4.9]. Since X is locally isomorphic to $W \times \mathbb{P}^{261}$ for open subsets $W \subset S$, X is smooth and $\dim X = 2 + 261 = 263$. Since S and the fibers of p are connected, X is connected. Since a smooth projective variety is Kähler [GH78, Chapter 0, §7], X is a connected compact Kähler manifold.

By Definition-Lemma 6.1 with $Z = S$, we have $T_X = T_{X/S} \oplus H_\nabla$, and dp restricts to an isomorphism $H_\nabla \to p^*T_S$. Let $V_1 := T_{X/S}$ and $V_2 := H_\nabla$. Then $\mathrm{rank}\, V_1 = 261$ and $\mathrm{rank}\, V_2 = \mathrm{rank}\, T_S = 2$. Since V_1 is the kernel of dp, it is the tangent bundle of the fibers of p, and so V_1 is integrable. By Construction-Proposition 4.5(2), for any $z \in S$ and $\xi, \zeta \in T_{S,z}$,

$$R_\nabla(\xi, \zeta) = \begin{pmatrix} 0 & \varphi(\xi, \zeta) \\ 0 & 0 \end{pmatrix}, \qquad R_\nabla(\xi, \zeta)^2 = 0,$$

and since $R_\nabla \neq 0$, there exist $z \in S$ and $\xi, \zeta \in T_{S,z}$ such that $R_\nabla(\xi, \zeta) \neq 0$. If $R_\nabla(\xi, \zeta) = \lambda\, \mathrm{id}_{E_z}$ for some $\lambda \in \mathbb{C}$, then $\lambda^2 = 0$, and so $R_\nabla(\xi, \zeta) = 0$, a contradiction. Thus $R_\nabla(\xi, \zeta)$ is not a scalar multiple of id_{E_z}, and Lemma 6.2 implies that V_2 is not integrable.

The projective space $\mathbb{P}^{261}$ is compact, connected, and simply connected [Hat02, Example 0.6 and Proposition 1.26], and $H^0(\mathbb{P}^{261}, \Omega^1_{\mathbb{P}^{261}}) = 0$ because any tangent vector of $\mathbb{P}^{261}$ is tangent to a line $\ell \subset \mathbb{P}^{261}$ and the restriction of a holomorphic 1-form to $\ell \cong \mathbb{P}^1$ is a section of $\Omega^1_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$, hence zero [Har77, II, Example 8.20.1] (holomorphic sections of $\mathcal{O}_{\mathbb{P}^1}(-2)$ are algebraic by GAGA

[Ser56, n° 12, Théorème 1], or vanish by Liouville). Moreover, S is connected with $\dim S = 2 > 0$, p is a holomorphic fiber bundle with fiber $\mathbb{P}^{261}$, and $q : \widetilde{S} \to S$ is a universal cover of S by Lemma 3.4. By Proposition 6.3 with $Z = S$, $\mathbb{P}^{261}$ in place of F, $H = H_\nabla$, and $g = q$, the projection

$$\pi : \widetilde{X} := X \times_S \widetilde{S} = \mathbb{P}(q^*E) \to X$$

is a universal cover of X.

Let Y_1, Y_2 be complex manifolds with $\dim Y_1 = 261$ and $\dim Y_2 = 2$, and let $\Phi : \widetilde{X} \to Y_1 \times Y_2$ be a biholomorphism. We show that such Y_1, Y_2, Φ do not exist. Since $\widetilde{X}$ is non-empty, Y_1 and Y_2 are non-empty. Thus $\mathrm{pr}_j \circ \Phi$ is surjective, and Y_j is connected as the image of the connected space $\widetilde{X}$, hence path-connected since Y_j is a manifold. Since $\widetilde{X}$ is simply connected, $\pi_1(Y_1) \times \pi_1(Y_2) \cong \pi_1(Y_1 \times Y_2)$ [Hat02, Proposition 1.12] is trivial, and so Y_1 and Y_2 are simply connected.

Let $\widetilde{p} : \widetilde{X} \to \widetilde{S}$ be the projection, let $z \in \widetilde{S}$, let $\widetilde{X}_z := \widetilde{p}^{-1}(z)$, and let u_z be the restriction of $\mathrm{pr}_2 \circ \Phi$ to $\widetilde{X}_z$. Then $\widetilde{X}_z \cong p^{-1}(q(z)) \cong \mathbb{P}^{261}$. Let $y \in u_z(\widetilde{X}_z)$. The fiber $u_z^{-1}(y)$ is identified with the intersection of the graph of u_z and $\widetilde{X}_z \times \{y\}$, which are closed complex submanifolds of dimension 261 of the complex manifold $\widetilde{X}_z \times Y_2$ of dimension 263. By the intersection inequality [GR84, Chapter 5, §3.1], the dimension of $u_z^{-1}(y)$ at any point of $u_z^{-1}(y)$ is at least $261 + 261 - 263 = 259$. By Chow's theorem [Ser56, n° 19, Proposition 13], $u_z^{-1}(y)$ is a closed algebraic subset of $\widetilde{X}_z \cong \mathbb{P}^{261}$, and by [Ser56, n° 6, Corollaire 3] and the union formula [GR84, Chapter 5, §3.1], the dimension of some irreducible component of $u_z^{-1}(y)$ is at least 259. Since $259 + 259 \geq 261$, any two closed subvarieties of $\mathbb{P}^{261}$ of dimension at least 259 intersect [Har77, I, Theorem 7.2]. As the fibers of u_z over distinct points of Y_2 are disjoint, $u_z(\widetilde{X}_z)$ consists of one point.

Thus $\mathrm{pr}_2 \circ \Phi$ is constant on any fiber of $\widetilde{p}$, and $\dim Y_2 = 2 = \dim S$. By the last statement of Proposition 6.3 under (b), applied with $Z = S$, $\mathbb{P}^{261}$ in place of F, $H = H_\nabla$, and $g = q$, there exists a biholomorphism $\widetilde{X} \to \mathbb{P}^{261} \times \widetilde{S}$ over $\widetilde{S}$, that is, $\mathbb{P}(q^*E) \cong \widetilde{S} \times \mathbb{P}^{261}$ over $\widetilde{S}$. This contradicts Theorem 1.4(5), which holds for E by Proposition 5.2. Hence X, V_1, and V_2 satisfy the statement of Corollary 1.5.

Since X is a connected compact Kähler manifold and $T_X = V_1 \oplus V_2$ with $\operatorname{rank} V_1, \operatorname{rank} V_2 > 0$, X, V_1, and V_2 satisfy the hypotheses of Conjecture 1.3 and of [Hör08, Conjecture 1.6]. By the argument above, there exist no complex manifolds Y_1, Y_2 with $\dim Y_j = \operatorname{rank} V_j$ for $j = 1, 2$ together with a biholomorphism $\widetilde{X} \to Y_1 \times Y_2$, so the first part of Conjecture 1.3 fails for X, V_1, and V_2. Since isomorphisms of vector bundles $\pi^* V_j \cong \Phi^* \mathrm{pr}_j^* T_{Y_j}$ imply $\dim Y_j = \operatorname{rank} V_j$, the first part of [Hör08, Conjecture 1.6] fails as well. Therefore, Conjecture 1.3 and [Hör08, Conjecture 1.6] are false. □

Remark 7.1. Let E and ∇ be the holomorphic vector bundle and the holomorphic connection given by Construction-Proposition 4.5, let $X = \mathbb{P}(E)$, and let $p : X \to S$ be the projection. Then X is a smooth projective variety (Corollary 1.5), p is a $\mathbb{P}^{261}$-bundle over a smooth projective surface, and the decomposition $T_X = T_{X/S} \oplus H_\nabla$ splits the relative tangent sequence

$$0 \to T_{X/S} \to T_X \to p^* T_S \to 0.$$

The universal cover of X is $X \times_S \widetilde{S} = \mathbb{P}(q^*E)$ (Lemma 3.4 and Proposition 6.3), and by Corollary 1.5 it is not biholomorphic to $\widetilde{S} \times \mathbb{P}^{261}$, since $\dim \widetilde{S} = 2$. On the other hand, [CP02, Theorem 2.7(1)] asserts that if X is a projective manifold with a $\mathbb{P}^r$-bundle structure $X \to Y$ whose relative tangent sequence splits, then the universal cover of X is isomorphic to $\widetilde{Y} \times \mathbb{P}^r$. Thus the conclusion of [CP02, Theorem 2.7(1)] does not hold for p. The proof in [CP02] constructs an isomorphism over $\widetilde{Y}$, and it uses [CP02, Proposition 2.6], which asserts that if G is a vector bundle of rank $r + 1$ on an irreducible reduced compact curve such that the relative cotangent sequence of $\mathbb{P}(G)$ over the curve splits, then the $\mathbb{Q}$-twisted bundle $G \otimes (\det G)^{1/(r+1)}$ is numerically

flat. Moreover, [CP02, Theorem 2.7(3)] asserts that if $X = \mathbb{P}(V)$ under the hypotheses of [CP02, Theorem 2.7], then $V \otimes (\det V)^{-1/(r+1)}$ is numerically flat. Both [CP02, Proposition 2.6] and [CP02, Theorem 2.7(3)] fail for the bundle E_0 on C: the relative tangent sequence of $\mathbb{P}(E_0) \to C$ splits by Lemma 2.5 and Definition-Lemma 6.1, hence so does the dual sequence (the relative cotangent sequence), and $\det E_0 \cong \mathcal{O}_C$, so they would imply that E_0 is numerically flat, that is, both E_0 and $E_0^\vee$ are nef. (In [CP02, Definition 2.1], numerical flatness of E means that the $\mathbb{Q}$-bundle $E \otimes (\det E^\vee)^{1/\operatorname{rank} E}$ is nef. Since $\operatorname{rank} E_0 = 2$ and $\det E_0 \cong \mathcal{O}_C$, $E_0^\vee \cong E_0$, so for E_0 numerical flatness in the sense of [CP02, Definition 2.1], and the convention for $\mathbb{P}(E_0)$, both reduce to the nefness of E_0.) However, E_0 is not nef, since it has the quotient $\mathcal{L}^{-1}$ with $\deg \mathcal{L}^{-1} = -1 < 0$ (cf. [Bis95, Remark 3.7(iii)] and [Mül25, Example 5.5]). The proof of [CP02, Proposition 2.6] deduces nefness from flatness [CP02, p. 623], while E_0 admits a flat holomorphic connection (Lemma 2.5) and is not nef. The integrable case [CP02, Theorem 2.7(2)] follows from [CP02, Proposition 2.5] and is not affected.

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